SPACE VEHICLE SAFETY CONSORTIUM (SVSC) | OFFICE OF DOCTRINE AND STANDARDS

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SPACE CAR SAFETY DOCTRINE

— Unified Edition —

Integrating Psychological Simulation, Behavioral Scoring, Safety Architecture,
Manufacturer Compliance, and Policy Framework

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Executive Summary

The **Space Car Safety Doctrine — Unified Edition** (hereinafter "the Doctrine") is the foundational regulatory and engineering reference governing the behavioral safety of autonomous and semi-autonomous Space Cars operating across terrestrial roadways, low-atmosphere airspace (below 15,000 feet above ground level), and transitional launch corridors. Issued by the Space Vehicle Safety Consortium (SVSC) and effective January 1, 2026, this Doctrine constitutes a binding standard for all manufacturers, operators, fleet managers, regulatory agencies, and Pilot-

Passengers (PPs) engaging with Space Car technologies within jurisdictions adopting the SVSC framework.

The Doctrine addresses a fundamental challenge that no prior vehicular safety standard has confronted at scale: the management of *human behavioral and psychological risk* within a propulsion environment that spans the terrestrial and near-orbital domains. Unlike traditional automotive or aviation safety frameworks, which focus primarily on mechanical integrity and environmental hazard, Space Car operations introduce a third category of risk — the psychological state of the individual exercising control within, or sharing authority with, an autonomous system. Behavioral volatility in a terrestrial automobile is dangerous. Behavioral volatility in a vehicle capable of transitioning from road to low-altitude flight is potentially catastrophic at a systemic level.

To address this challenge, the Doctrine establishes three interconnected frameworks:

1. **The Three-Path Psychological Simulation Engine** — a real-time behavioral assessment and path-assignment system that continuously classifies the PP's emotional and cognitive state and routes the vehicle through one of three operational courses: the *Peaceful Course* (nominal, low-risk), the *Naughty Course* (elevated-risk, active intervention), or the *Whisper Course* (covert de-escalation, embedded within the Naughty Course).
2. **The Behavioral Risk Index (BRI)** — a composite scoring system (0–100) computed from six weighted sub-indices spanning physiological, kinematic, historical, environmental, cognitive, and social signals. The BRI drives all automated safety responses and threshold triggers.
3. **The Policy and Compliance Framework** — a comprehensive set of manufacturer specifications, training requirements, liability allocations, regulatory integrations, and incident reporting obligations that governs the entire Space Car ecosystem from design to retirement.

Key provisions of this Doctrine include mandatory Psychometric Sensor Array (PSA) hardware in all Space Car classes; a seven-state Emotional State Classification (ESC) system with dedicated protocols for vulnerable and crisis states; a full training curriculum for Pilot-Passengers, fleet operators, and first responders; software ethics certification requirements including an independent review of the Whisper Course implementation; and a clear liability allocation matrix protecting compliant manufacturers while preserving full PP rights to data transparency and human review of automated decisions.

■ Regulatory Note

This Doctrine does not supersede existing federal aviation, transportation, or communications law. It operates as a supplementary behavioral safety standard layered upon FAA UAM regulations, NHTSA autonomous vehicle guidelines, FCC spectrum allocations, ISO 26262 functional safety requirements, and SAE J3016 autonomy level definitions. Where conflict arises, federal statute prevails and the SVSC Technical Committee must be notified within 30 days for doctrine reconciliation proceedings.

The SVSC calls upon all Space Car manufacturers, government partners, and PP advocates to treat this Doctrine not as a ceiling of ambition but as a floor of obligation. Safety in the Space Car era is not a feature to be marketed — it is the operating substrate upon which every journey, at every altitude, must be built.

Part I

Foundational Doctrine

1. Purpose, Scope, and Governing Philosophy

1.1 Mission Statement

The Space Vehicle Safety Consortium affirms the following as the foundational axiom of this Doctrine:

> Safety is not a feature — it is the operating substrate of every Space Car interaction.
>
> *Every sensor, every algorithm, every policy provision, and every training module contained in this Doctrine serves a single master purpose: ensuring that every person who enters, operates, or is proximate to a Space Car arrives at their destination — or is safely and with dignity diverted from it — without injury, psychological harm, or systemic risk to the broader public.*

This mission statement is not aspirational language. It is an operative directive. Every manufacturer decision, every software architecture choice, and every regulatory determination made under the authority of this Doctrine shall be evaluated against this standard. Where a design, protocol, or policy fails to treat safety as the substrate — where it treats safety as an add-on, a trade-off, or a marketing differentiator — it is non-compliant with this Doctrine by definition, regardless of whether it satisfies specific numerical thresholds.

1.2 Scope

This Doctrine applies to all Space Cars and all entities engaged in their design, manufacture, sale, operation, maintenance, regulation, or use, in the following operational environments:

- **Terrestrial Roadway Operations:** All operations on paved, graded, or designated surface roadways, including parking structures, private roads, toll facilities, and emergency access routes.
- **Low-Altitude Airspace Operations:** All operations at altitudes from surface departure to 15,000 feet above ground level (AGL), including hover, transit, and descent phases. This range encompasses Class G and portions of Class E airspace as defined by the FAA, and all Urban Air Mobility (UAM) corridors designated under FAA UAM Integration Framework.
- **Transitional Launch Corridors:** Defined segments of airspace and surface infrastructure where a Space Car transitions between terrestrial and airborne operation modes, including vertical takeoff and landing (VTOL) pads, launch ramps, and controlled transition zones.

This Doctrine applies irrespective of:

- Vehicle ownership type (private, commercial, fleet, governmental, military non-combat);
- Journey purpose (personal, commercial, emergency, demonstration);
- Automation level (fully autonomous Class IV through human-piloted Class I as defined in Section 2.2);
- Pilot-Passenger age, status, or certification level, except where specific provisions create differentiated requirements.

ⓘ **Scope Note**

Military combat Space Car operations, orbital-insertion vehicles operating above 15,000 ft AGL, and experimental prototype vehicles operating under FAA Certificate of Waiver or Authorization (COA) are

exempt

from this Doctrine unless voluntarily adopted by the operating authority. Manufacturers producing dual-use (civilian and military) vehicles must implement this Doctrine in full for all civilian variants.

1.3 Governing Philosophy: The Triad of Safety

The SVSC Doctrine is structured around a three-pillar philosophy of safety that recognizes the multi-dimensional nature of risk in Space Car operations. No single dimension of safety is sufficient in isolation; all three must be simultaneously maintained for an operation to be considered safe under this Doctrine.

Safety Pillar	Definition	Primary Instruments	Failure Mode
Physical Safety	The prevention of bodily injury to the PP, passengers, third parties, and property through mechanical integrity, collision avoidance, and structural systems.	ISO 26262 compliance, Hard Lock, Safe Harbor navigation, redundant sensor arrays	Mechanical failure, collision, structural breach, unsafe landing
Psychological Safety	The protection of the PP's mental and emotional state during operation, ensuring cognitive capacity is not overwhelmed and that distress does not cascade into physical danger.	BRI system, ESC classification, Peaceful/Naughty/Whisper Courses, Individual Psychological Profile	Panic response, impaired judgment, deliberate self-harm, emotional override of system warnings

Safety Pillar	Definition	Primary Instruments	Failure Mode
Systemic Safety	The preservation of safety for the broader transport ecosystem — airspace integrity, traffic flow, emergency service access, and societal trust in Space Car technology.	Geofencing, incident reporting, BRI 90+ protocols, regulatory integration, fleet-level compliance monitoring	Systemic airspace conflict, public endangerment, regulatory non-compliance, erosion of public trust

The Triad of Safety is not hierarchical — Physical Safety does not automatically supersede Psychological Safety, nor does Systemic Safety subsume individual welfare. The three pillars exist in dynamic tension, and the architecture of this Doctrine — particularly the BRI system and the Three-Course Framework — is specifically designed to navigate that tension in real time, at the level of individual journeys, without requiring human intervention for every decision.

1.4 Doctrine Hierarchy

This Doctrine integrates with the following existing regulatory and standards frameworks. The hierarchy below defines precedence in the event of conflict:

4. **Federal Statute and Constitutional Law** — Supreme authority; this Doctrine does not and cannot supersede federal law.
5. **FAA Regulations (14 CFR)** — Primary authority for all airspace operations; Space Car operations in any airborne phase are subject to FAA jurisdiction.
6. **NHTSA Regulations (49 CFR)** — Primary authority for all surface vehicle operations; NHTSA Federal Motor Vehicle Safety Standards (FMVSS) apply to all terrestrial Space Car operations.
7. **FCC Part 15 and Relevant Spectrum Rules** — Govern all RF-based sensor and communication systems used in the Psychometric Sensor Array (PSA).

8. **ISO 26262:2018 (Functional Safety)** — Applied to all safety-critical electronic systems; Space Cars must achieve ASIL-D (Automotive Safety Integrity Level D) for BRI computation and Hard Lock systems.
9. **SAE J3016 (Autonomy Levels)** — Defines the six levels of driving automation referenced throughout this Doctrine for classification of Space Car operational modes.
10. **This Doctrine (SVSC-DOC-2026-001)** — Applies specifically to behavioral safety dimensions not addressed by the frameworks above, and supplements those frameworks with psychological safety and course management provisions.

> ■ **Regulatory Note — Integration Obligation**
>
> Manufacturers and operators must maintain documented evidence of compliance with all frameworks listed in Section 1.4. The SVSC does not perform FAA, NHTSA, FCC, or ISO certification on behalf of manufacturers. SVSC certification is a supplementary credential that does not substitute for primary regulatory approvals. Manufacturers who obtain SVSC certification before completing primary regulatory approvals are not authorized to commercially operate Space Cars.

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2. Definitions and Taxonomy

2.1 Glossary of Terms

The following terms are defined for purposes of this Doctrine. All capitalized uses of these terms throughout the document refer to these definitions unless explicitly stated otherwise.

Term	Definition
Space Car	A multi-modal autonomous or semi-autonomous vehicle capable of operating in at least two of the following environments: terrestrial roadway, low-altitude airspace (below 15,000 ft AGL), or transitional launch corridor. Space Cars must integrate a Psychometric Sensor Array and comply with this Doctrine.
Pilot-Passenger (PP)	The primary registered human occupant of a Space Car who is authorized to issue navigational commands, adjust journey parameters, and invoke manual override protocols. A PP bears primary behavioral responsibility for interactions with the vehicle's safety systems.
Behavioral Risk Index (BRI)	The composite real-time score (0–100) representing the current behavioral risk level of a PP, calculated from six weighted sub-indices. BRI drives all automated course assignments, threshold triggers, and safety interventions.
Emotional State Classification (ESC)	The seven-state categorical classification of a PP's detected emotional and cognitive condition, ranging from ESC-1 (Serene) to ESC-7 (Crisis). The ESC complements the BRI by providing qualitative context for automated responses.
Course Path	The operational mode assigned to a Space Car's behavioral management system based on BRI and ESC evaluation. Three Course Paths exist: Peaceful Course, Naughty Course, and Whisper Course.
Whisper Protocol	The specific set of covert, non-declarative interventions applied during the Whisper Course, designed to

Term	Definition
	reduce PP arousal and BRI without the PP being aware of the course reassignment. See Section 6.3 for full mechanics.
Naughty-Course Flag	A discrete categorical trigger event (as distinct from a BRI threshold crossing) that independently activates the Naughty Course regardless of the current BRI score. See Section 5.2 for the full list of flag conditions.
Peaceful Baseline	The individually calibrated BRI and biometric baseline established for each PP during the first 90 days of Space Car ownership. The Peaceful Baseline serves as the reference point against which deviations are measured.
Threshold Override	An automated system action triggered at a specific BRI threshold that cannot be countermanded by PP input without biometric re-authorization, as specified in Section 8.1.
Soft Lock	The operational state triggered at BRI 70–74 in which the Space Car reduces maximum speed to 25 mph, restricts navigation destinations to Safe Harbor locations, and initiates emergency services pre-staging. PP can request de-escalation review. See Section 5.3, Phase 3.
Hard Lock	The automated system state triggered at BRI 75 in which the Space Car assumes full autonomous control, dispatches emergency services, and restricts PP input to emergency communication only. See Section 8.2.
Geofence	A defined virtual geographic boundary enforced by the Space Car's navigation system that constrains or modifies vehicle behavior within or at the boundary of a specified area. Geofences may be static (permanent zone designations) or dynamic (situationally generated based on BRI or incident events).
Psychometric Sensor Array (PSA)	The integrated suite of hardware

Term	Definition
	sensors embedded in the Space Car cabin responsible for collecting biometric, kinematic, visual, and acoustic data used in BRI and ESC computation. See Section 10.2 for hardware specifications.
Safe Harbor	A designated, pre-approved landing or parking location equipped with emergency services access, structural safety verification, and perimeter security, to which the Space Car automatically navigates upon Hard Lock engagement or BRI 90+ events.
Individual Psychological Profile (IPP)	The persistent, privacy-protected data record maintained for each PP that stores Peaceful Baseline data, preferred de-escalation modalities, historical trigger patterns, and BRI history for adaptive calibration purposes.
Single-Event Override Flag (SEOF)	A discrete detection event that immediately triggers Hard Lock regardless of current BRI score. SEOFs bypass all BRI thresholds and course path logic. See Section 8.3 for the complete list.
Physiological Sub-Index (PSI)	The BRI sub-component (weight: 30%) derived from physiological signals including heart rate variability, galvanic skin response, micro-expression analysis, and vocal stress patterns.
Kinematic Sub-Index (KSI)	The BRI sub-component (weight: 20%) derived from movement signals including gesture velocity, steering input force, and body position shifts detected by seat and cabin sensors.
Historical Risk Sub-Index (HRI)	The BRI sub-component (weight: 15%) derived from the PP's 30-day average BRI, prior Naughty Course activations, and accumulated flag history.
Environmental Sub-Index (ESI)	The BRI sub-component (weight: 10%) derived from contextual environmental factors including traffic density, weather severity, time of day, and current geofence zone designation.

Term	Definition
Cognitive Load Sub-Index (CLI)	The BRI sub-component (weight: 15%) derived from cognitive performance signals including response latency to system prompts and dual-task performance metrics.
Social Dynamics Sub-Index (SDI)	The BRI sub-component (weight: 10%) derived from cabin social context including passenger count, detected verbal conflict, and presence of child or other vulnerable occupant flags.
Binaural Relaxation Frequencies	Auditory stimuli — specifically audio tracks or tonal overlays at frequencies associated with reduced physiological arousal — deployed covertly during the Whisper Protocol as a calming intervention.
Micro-Acceleration Smoothing	A Whisper Protocol intervention in which the Space Car's propulsion control system imperceptibly reduces acceleration variance to decrease physiological arousal in the PP without triggering conscious awareness of the change.
BRI Decay	The natural or intervention-assisted reduction of BRI over time in the absence of new triggering events. Natural decay rate is 2 BRI points per minute under calm conditions.
Compliant Fleet Operator	A commercial or governmental entity operating a fleet of Space Cars that has completed the Fleet Operator Certification program, maintains documented SVSC compliance records, and fulfills quarterly reporting obligations.
SVSC Technical Review Board (TRB)	The standing technical committee of the Space Vehicle Safety Consortium responsible for certifying BRI algorithms, reviewing Whisper Course implementations, approving OTA software updates, and adjudicating manufacturer compliance disputes.
Over-the-Air (OTA) Update	A remote software update delivered to a Space Car's onboard systems via secure wireless transmission. Safety-

Term	Definition
	critical OTA updates require SVSC TRB pre-approval under Section 14.3.
Agitation-on-Notification (AoN)	The detected behavioral pattern in which explicit notification of course reassignment or system intervention increases rather than decreases the PP's BRI. AoN is a primary activation criterion for the Whisper Course.

2.2 Vehicle Classification Tiers

All Space Cars are assigned to one of three classification tiers based on their operational envelope. Classification determines applicable hardware requirements, PP training obligations, and regulatory filing requirements.

Class	Designation	Operational Domain	Max Altitude	PP Certification Required	PSA Tier
Class I	Terrestrial-Only	Surface roadways exclusively; no airborne capability	N/A (surface only)	Standard (Modules 1–4)	PSA-1 (standard suite)
Class II	Low-Altitude	Surface roadways plus airborne operations up to 15,000 ft AGL	15,000 ft AGL	Advanced (Modules 1–5; 85% pass threshold)	PSA-2 (enhanced suite with redundant backups)
Class III	Transitional Orbital	Surface, low-altitude, and transitional launch corridors;	As authorized by FAA COA; doctrine applies through	Full Certification (Modules 1–5; additional orbital-interface	PSA-3 (full-redundancy suite with external environment sensors)

Class	Designation	Operational Domain	Max Altitude	PP Certification Required	PSA Tier
		may interface with orbital approach corridors above 15,000 ft	15,000 ft	training; re-certification annually)	

⚠️ **Safety Critical — Class Modification**

Modifications that extend a Space Car's operational envelope beyond its classified tier (e.g., installing propulsion hardware on a Class I vehicle to achieve airborne capability) without completing reclassification to Class II constitute a violation of this Doctrine and applicable FAA regulations. Such vehicles are immediately non-compliant and must be grounded pending full Class II certification review.

Part II

Three-Path Psychological Simulation

3. The Three-Course Framework

3.1 Overview

The Space Car's behavioral management architecture is built upon a real-time psychological simulation engine — referred to throughout this Doctrine as the **Three-Course Framework** — that continuously evaluates the Pilot-Passenger's emotional, cognitive, and behavioral state and assigns one of three operational Course Paths. The Three-Course Framework is not a reactive safety net; it is the primary operational architecture within which every Space Car journey is managed from ignition to arrival.

The simulation engine runs in a continuous closed loop with a maximum cycle time of 500 milliseconds, meaning the BRI is recalculated and the Course Path re-evaluated at least twice per second under normal computational load. Under elevated-risk conditions (BRI 50+), the cycle time reduces to 200 milliseconds to ensure rapid response to escalating behavioral signals.

The three Course Paths are:

- **PATH 1 — Peaceful Course:** Nominal operational state. Full navigation autonomy. Passive monitoring. Comfort-optimized journey environment.
- **PATH 2 — Naughty Course:** Elevated-risk state. Active intervention cascade. Restricted navigation options. Emergency services pre-staging at higher phases.
- **PATH 3 — Whisper Course:** Covert de-escalation pathway embedded within the Naughty Course. Transparent to the PP. Applied when direct notification would worsen the PP's state.

3.2 Path Selection Logic

The Course Path assignment process follows a structured decision tree executed at each simulation cycle. The stages are as follows:

11. **Initial State Assessment:** Upon journey initiation, the Space Car performs a baseline biometric capture (30-second pre-departure scan) and retrieves the PP's Individual Psychological Profile (IPP). An initial BRI score is computed

from available sub-indices. If the PP has no IPP (first use), population-average baselines are used until a personal Peaceful Baseline is established.

12. **Dynamic Sensor Read:** At each 500ms cycle (200ms if BRI ≥ 50), the PSA captures a real-time multimodal data snapshot across all six sub-indices. Raw sensor data is pre-processed by the onboard sensor fusion module before BRI computation.
13. **Threshold Comparison:** The computed BRI is compared against all defined thresholds (Section 8.2) and checked against the Single-Event Override Flag (SEOF) registry (Section 8.3). ESC is simultaneously classified (Section 9.1).
14. **Course Assignment:** Based on BRI score and ESC state, the simulation engine assigns the current Course Path. Assignment decisions are logged with timestamp and sensor snapshot.
15. **Continuous Monitoring Loop:** The assigned Course Path's specific behavioral protocols and system responses are engaged. The loop restarts at Step 2. Course Path transitions are logged as discrete events with full BRI/ESC state captured at the moment of transition.

⚠️ Safety Critical — Simulation Engine Continuity

The Three-Course simulation engine must maintain continuous operation for the duration of every journey. System interruption of the simulation engine — including during OTA updates, sensor recalibration cycles, and vehicle mode transitions between terrestrial and airborne operation — is prohibited. A redundant co-processor must maintain BRI computation during any primary processor interruption. Failure of both primary and backup simulation processors must trigger immediate Soft Lock engagement.

4. PATH 1 — The Peaceful Course

4.1 Definition

The **Peaceful Course** is the nominal operational state of the Space Car, representing the journey experience as intended under conditions of PP behavioral stability and low environmental risk. The Peaceful Course is assigned when the Behavioral Risk Index is below 25 AND the Emotional State Classification is one of the following: ESC-1 (Serene), ESC-2 (Focused), ESC-3 (Mildly Elevated). The Peaceful Course is the default state at journey initiation absent any Pre-Departure Override Condition (PDOC).

The Peaceful Course is not a passive state from the vehicle's perspective. It is an actively maintained environment of safety and comfort, continuously managed to preserve the PP's favorable behavioral state and reduce the likelihood of BRI escalation. Think of the Peaceful Course not as the absence of safety action, but as the highest form of preventive safety architecture — ensuring the conditions that make escalation unnecessary.

4.2 System Behaviors in Peaceful Course

When assigned to the Peaceful Course, the Space Car engages the following operational behaviors:

- **Full Navigation Autonomy:** All registered destination categories are enabled, including personal addresses, commercial destinations, restricted zones requiring PP-level clearance, and air corridor entry points (Class II and III vehicles). No geographic restrictions are applied.
- **Ambient Environmental Enhancement:** The Space Car's Environmental Conditioning System (ECS) applies the PP's preferred ambient sound profile, lighting spectrum, temperature setting, and seat configuration as stored in the IPP. Default profiles apply for first-use PPs.

- **Soothing Sound Profiles:** Audio environment is optimized for the PP's stored preferences. If no preference is stored, the system defaults to a neutral, low-stimulation ambient soundtrack calibrated to reduce physiological arousal and support sustained ESC-1 or ESC-2 states.
- **Optimized Lighting:** Cabin lighting defaults to a warm, moderate-intensity spectrum during daytime and a soft, blue-spectrum-reduced profile during evening and nighttime operations, consistent with research on circadian regulation and cognitive calm.
- **Gentle Acceleration Curves:** The propulsion system applies smooth acceleration and deceleration profiles that minimize kinematic stress on passengers. Sharp acceleration curves associated with high-performance driving are available only upon explicit PP request and are automatically moderated if KSI readings indicate kinematic stress response.
- **Proactive Route Optimization:** The navigation system continuously evaluates routing options for the lowest-stress profile, factoring in traffic density, weather, airspace congestion (Class II/III), and time-of-day patterns. The PP is offered alternative routes if a superior ESI (lower environmental stress) option is available.

4.3 Pilot-Passenger Experience Design

The Peaceful Course experience is individualized rather than standardized. The Space Car accumulates PP preference data over time and builds a personalized journey profile within the IPP. The following are the core dimensions of Peaceful Course experience personalization:

- **Sensory Environment Calibration:** Temperature, humidity, fragrance (where cabin scent systems are installed), ambient sound, and lighting are adjusted to within the PP's documented optimal range. Adjustments occur gradually (over 3–5 minutes) to avoid abrupt environmental shifts.

- **Comfort Preference Learning:** The system records PP adjustments to system-generated defaults. If a PP consistently overrides the default lighting profile to a warmer tone, the system updates the IPP after three consistent overrides. Preference learning is constrained — learned preferences cannot override safety-critical environmental settings.
- **Personalized Journey Profiling:** The system learns journey rhythm preferences (e.g., PP prefers silence for the first 10 minutes of morning journeys; PP engages with news content during commute hours). These patterns are incorporated into journey initiation defaults.
- **Passenger-Adaptive Mode:** When additional occupants are detected, the system adapts the environmental profile to a shared-comfort configuration and updates the SDI component of the BRI. If child occupants are detected, additional safety protocols are silently engaged (child-appropriate audio, reduced speed envelope, enhanced SDI monitoring).

4.4 Monitoring Protocol

In the Peaceful Course, the PSA operates in a passive, low-intensity monitoring mode to preserve PP privacy while maintaining the baseline vigilance necessary for rapid detection of BRI escalation. The following monitoring protocol applies:

- **PSA Sampling Rate:** Passive biometric sampling every 30 seconds. Kinematic sampling every 5 seconds (reduced from active-mode 500ms cycle).
- **No Active Intervention:** No calming interventions, route restrictions, or PP notifications are issued unless BRI crosses 25 or an SEOF is detected.
- **Trend Monitoring:** The simulation engine monitors not only the absolute BRI value but its rate of change. A BRI that increases by more than 8 points within any 2-minute window in Peaceful Course triggers an escalation review cycle (PSA sampling increases to every 10 seconds) even if the absolute BRI remains below 25.

- **Environmental Pre-scan:** In Class II and III operations, the system performs a 5-minute airspace ESI pre-scan before altitude transition to detect conditions (turbulence forecast, traffic density, weather approach) that may increase ESI and thus increase BRI during transition. PPs are offered optional delay or route modification if the ESI pre-scan identifies elevated environmental risk.

4.5 Peaceful Course Scoring Rewards

To incentivize and reinforce the behavioral conditions associated with safe Space Car operation, the Doctrine provides for positive scoring rewards within the Peaceful Course framework. These rewards contribute to the PP's long-term safety record and may be applied by manufacturers or fleet operators toward insurance scoring, access privileges, or loyalty programs, subject to PP data consent provisions.

- **BRI Reduction Bonus:** For every continuous 10-minute interval in which the PP remains in Peaceful Course with BRI below 15 (ESC-1 Serene), a BRI reduction bonus of −5 points is applied to the Historical Risk Sub-Index (HRI). This bonus is cumulative across journeys and represents the positive behavioral history that lowers the PP's long-term BRI baseline.
- **Consecutive Journey Credit:** A PP who completes 10 consecutive journeys entirely within the Peaceful Course earns a Consecutive Journey Credit recorded in the IPP. After 30 such journeys, the PP achieves a "Platinum Peaceful" status that modifies the BRI threshold for Naughty Course activation from 50 to 55 for that individual, reflecting demonstrated behavioral reliability.
- **Loyalty Scoring Contribution:** With PP consent, Peaceful Course journey data — specifically the duration, depth (average BRI), and frequency of Peaceful Course operation — may be shared with participating insurance providers and fleet management platforms for premium calculation and loyalty benefit purposes. No raw biometric data may be shared under any loyalty program arrangement.

5. PATH 2 — The Naughty Course

5.1 Definition

The **Naughty Course** is the elevated-risk operational state of the Space Car, activated when the PP's Behavioral Risk Index and/or behavioral signals indicate that the conditions of safe autonomous operation are no longer fully met in the Peaceful Course framework. The Naughty Course is not a punitive state — it is a protective state, designed to maintain safety for the PP, passengers, and the public while preserving dignity and applying graduated, proportionate interventions rather than immediate restriction.

The Naughty Course is triggered under either of two conditions:

- **BRI Threshold Trigger:** BRI reaches or exceeds 50.
- **Behavioral Flag Trigger:** Any single behavioral flag from the Naughty-Course Flag registry is detected, regardless of the current BRI score.

Once activated, the Naughty Course operates through a three-phase cascade of escalating interventions, each phase triggered by BRI range and ESC classification. The Naughty Course is exited either through successful de-escalation (BRI drops below 50) or through escalation to Hard Lock (BRI 75+).

5.2 Trigger Conditions

Trigger Type	Condition	BRI Required	Immediate Effect
BRI Threshold	Composite BRI score reaches or exceeds 50	BRI ≥ 50	Naughty Course Phase 1 engaged

Trigger Type	Condition	BRI Required	Immediate Effect
Behavioral Flag	Aggressive verbal input detected (threat vocabulary, sustained high-volume vocalization)	Any BRI	Naughty Course Phase 2 minimum; BRI incremented by +10
Behavioral Flag	Biometric stress indicators — elevated cortisol proxy signal, heart rate above 140 bpm sustained 60+ seconds	Any BRI	Naughty Course Phase 1; BRI incremented by +5
Behavioral Flag	Erratic manual override gestures — steering input force exceeding 2x normal operating threshold	Any BRI	Naughty Course Phase 1; BRI incremented by +7
Behavioral Flag	Proximity-aggression toward other vehicles — aggressive proximity maneuvers detected by external sensor array	Any BRI	Naughty Course Phase 2; BRI incremented by +12
Behavioral Flag	Speed-demand escalation — PP requests speed exceeding posted limit or safe corridor speed by more than 20%	Any BRI	Naughty Course Phase 1; request denied; BRI incremented by +8
Behavioral Flag	Substance-impairment signal from PSA — reaction latency, gaze tracking deviation, vocal slurring pattern exceeding	Any BRI	Naughty Course Phase 2 minimum; emergency contact pre-alert triggered

Trigger Type	Condition	BRI Required	Immediate Effect
	impairment threshold		

5.3 System Response Cascade

5.3.1 Phase 1: Gentle Ambient Recalibration (BRI 50–59)

Phase 1 represents the first tier of active intervention. The goal of Phase 1 is to create conditions conducive to BRI reduction without restricting the PP's navigational freedom or generating alarm. Interventions are subtle, environmental, and non-declarative. The PP is not informed of the Naughty Course assignment at Phase 1 unless they explicitly query the system about their current course status.

- **Ambient Recalibration:** The ECS transitions the cabin environment toward the PP's documented de-escalation profile. Ambient sound shifts toward lower-frequency, lower-tempo content. Lighting shifts toward a cooler, lower-intensity spectrum at a rate of 2% per minute. Temperature is adjusted toward the PP's documented calming temperature range.
- **Speed Cap:** Maximum speed is capped at posted speed limit plus 10%. Speed cap is applied gradually (1 mph per 5 seconds) to avoid abrupt deceleration that may increase kinematic stress.
- **Route Optimization:** The navigation system actively routes away from high-congestion zones, construction areas, and high-incident corridors. If the PP's requested destination requires transit through a high-ESI zone, the system offers an alternative route with lower ESI.
- **Simulation Cycle Acceleration:** PSA sampling rate increases to every 10 seconds. BRI recalculation cycle decreases to 200ms. The system enters heightened monitoring mode for the duration of Phase 1.

5.3.2 Phase 2: Active Whisper Engagement (BRI 60–69)

Phase 2 intensifies system intervention and introduces destination restrictions and emergency contact pre-alerting. At BRI 60, the system also assesses ESC to determine whether Whisper Course activation is appropriate (see Section 6.2). If Whisper Course is not activated, Phase 2 proceeds with explicit (non-covert) intervention protocols.

- **Whisper Course Eligibility Check:** ESC is evaluated against Whisper Course activation criteria. If criteria are met, Whisper Course is activated in parallel with Phase 2 protocols. If not met, Phase 2 explicit protocols proceed.
- **Speed Cap:** Maximum speed is reduced to exactly the posted speed limit. No exceptions are permitted during Phase 2, including PP requests for speed increase.
- **Destination Restriction:** Navigation destination options are reduced to pre-approved safe categories: PP's registered home address, registered safe contacts' addresses, medical facilities, Space Car service centers, and designated Safe Harbor zones. Commercial, entertainment, and unregistered destinations are temporarily disabled.
- **Emergency Contact Pre-Alert:** The PP's designated emergency contact is sent a silent pre-alert notification: *"[PP Name]'s Space Car is currently managing an elevated-caution journey. No action required at this time. You will be contacted immediately if the situation escalates."* The PP is informed that an emergency contact pre-alert has been issued.
- **Air Corridor Suspension (Class II/III):** Any pending altitude transition is suspended. The vehicle completes the current segment at terrestrial altitude and does not initiate airborne operations until BRI drops below 50.

5.3.3 Phase 3: Soft Lock (BRI 70–74)

Phase 3 engages the Soft Lock state — the most restrictive non-Hard Lock condition. At this phase, the PP's behavioral risk is assessed as severe enough to require significant operational restriction, though Hard Lock has not yet been triggered.

- **Speed Reduction to 25 mph Maximum:** Vehicle speed is reduced to 25 mph maximum regardless of road type, posted limit, or PP request. Reduction occurs at a controlled rate of 2 mph per second.
- **Destination Locked to Nearest Safe Harbor:** Navigation is redirected to the nearest designated Safe Harbor zone. The PP is explicitly informed: *"Your Space Car is navigating to a Safe Harbor location for your safety. Emergency services have been pre-staged at this location. You may contact your emergency contact or emergency services directly at any time."*
- **Emergency Services Pre-Staging:** The Space Car transmits a Phase 3 pre-staging request to the nearest emergency services dispatch. Emergency services are notified of the vehicle's current GPS location, Safe Harbor destination, current BRI, and ESC classification. They are placed on standby — not dispatched — at this stage.
- **PP Notification of Status:** The PP is clearly informed of their current operational status, the reason for Soft Lock engagement in general terms, and the de-escalation pathway available to them (see Section 5.4).
- **Cabin Recording Activation:** Full cabin audio and video recording begins (encrypted, retained per Section 14.4) for regulatory and incident review purposes. PP is notified of recording activation.

5.4 Naughty Course De-escalation Protocol

The Naughty Course includes a structured, accessible pathway for PP de-escalation and return to the Peaceful Course. The de-escalation protocol is designed to be non-punitive, achievable through genuine behavioral change, and respectful of PP autonomy within the bounds of safety requirements.

- **BRI Natural Decay:** BRI naturally decays at 2 points per minute under calm conditions. In Phase 1, this decay is sufficient to return the PP to Peaceful Course within 25 minutes absent new triggering events.

- **Intervention-Assisted Decay:** If the PP engages constructively with available calming tools (voluntary guided breathing exercises offered on the dashboard display, voluntary calming audio content, voluntary system-assisted meditation prompt), BRI decay is augmented by an additional 1 point per minute per engaged intervention, up to a maximum of 5 additional points per minute.
- **Re-qualification Threshold:** A PP in Phase 2 or Phase 3 must sustain BRI below 45 for a minimum of 5 continuous minutes before being elevated back to Phase 1. A further 5 continuous minutes below 25 returns the PP to Peaceful Course. These re-qualification windows prevent rapid oscillation between course states.
- **Safe Harbor De-escalation Option:** A PP who has reached Phase 3 (Soft Lock) may voluntarily request to pause the journey at any Safe Harbor-equivalent location (parking area, rest stop). Upon confirmed stop and 10-minute calm monitoring interval, de-escalation review is initiated.

5.5 Naughty Course Logging and Reporting

All Naughty Course activations are automatically and comprehensively logged. Logging is non-optional and cannot be disabled by the PP or by any operator, manufacturer, or third-party instruction.

- **Automatic Encrypted Event Log:** Every Naughty Course event — from activation through de-escalation or escalation — generates an encrypted event log entry containing: timestamp, BRI trajectory (full 500ms-resolution BRI readings for the event duration), ESC classifications at each transition, specific trigger conditions, interventions applied, and outcome.
- **Local Retention:** Naughty Course logs are retained in local encrypted storage for a minimum of 90 days.
- **Regulatory Reporting Thresholds:** Phase 1 events are logged locally only. Phase 2 events generate a manufacturer report within 72 hours. Phase 3

events generate an SVSC report within 24 hours. See Section 18 for full incident classification and reporting requirements.

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6. PATH 3 — The Whisper Course

6.1 Definition

The **Whisper Course** is the covert de-escalation pathway embedded within the Naughty Course. It is the most ethically sensitive component of this Doctrine and is subject to the most stringent ethical constraints, certification requirements, and oversight provisions of any feature described herein. The Whisper Course does not exist as a separate operational state from the Naughty Course — it is a parallel intervention layer that operates within Naughty Course Phase 2 or Phase 3, transparently to the PP.

The defining characteristic of the Whisper Course is its transparency to the Pilot-Passenger: the system applies calming behavioral interventions without the PP being explicitly aware that a course reassignment has occurred and that targeted calming is underway. This approach is predicated on the empirically supported finding that, for a specific subset of individuals in a specific range of emotional states, explicit notification of safety intervention — *"You are being monitored for elevated behavioral risk"* — produces an Agitation-on-Notification response that increases BRI rather than decreasing it, thereby worsening rather than improving the safety situation.

▲ Ethical Constraint — Foundational Principle

The Whisper Course exists to protect the PP and the public through evidence-based calming, not to deceive, manipulate, or suppress PP agency. Every Whisper Protocol intervention is calibrated to create genuine physiological calm — the same interventions a skilled human caregiver might apply in an acute stress situation. The Whisper Course is a tool of care, not control. This distinction must be preserved at every level of implementation, certification, and operation.

6.2 Activation Criteria

The Whisper Course is activated when ALL of the following conditions are simultaneously met:

16. **BRI ≥ 60** (Naughty Course Phase 2 or Phase 3 is active).
17. **ESC flag of "Resistant to Direct Intervention" (RDI) OR "Agitation-on-Notification" (AoN)** is present. RDI is flagged when the PP has demonstrably rejected or escalated in response to three or more direct interventions within the current journey or the last 7 days of journeys. AoN is flagged when real-time biosensor data shows BRI increase following notification events in the current or most recent journey.
18. **Whisper Course is not contraindicated** by any active condition. Contraindications include: active SEOF, BRI 75+ (Hard Lock supersedes), PP's IPP contains a documented Whisper Course opt-out, or current ESC-7 (Crisis) status requires explicit emergency engagement rather than covert intervention.

When all criteria are met, the simulation engine silently activates the Whisper Course. No notification is provided to the PP. The Whisper Course activation is logged as a discrete event in the encrypted event log.

6.3 Whisper Protocol Mechanics

The Whisper Protocol consists of five classes of covert intervention, applied in combination and calibrated to the PP's individual IPP profile. All interventions are designed to operate at or below the threshold of conscious perception while producing measurable physiological effects consistent with reduced arousal.

6.3.1 Ambient Sound Shift

The cabin audio system introduces binaural relaxation frequencies — low-frequency audio overlays (typically in the 40–60 Hz delta/theta range) — gradually mixed into the existing audio content at an initial level of −30 dB below the dominant audio track. Over 5 minutes, the mix level is increased to −15 dB. The overlay is acoustically masked by the dominant content and is not consciously distinguishable from the existing audio experience. Where no audio content is playing, a low-level ambient sound layer (rain, distant ocean, subtle instrumental) is introduced at −20 dB starting volume, blended into cabin ambient noise.

6.3.2 Micro-Acceleration Smoothing

The Space Car's propulsion control system imperceptibly reduces acceleration variance — the moment-to-moment fluctuation in acceleration force — by 15% over 3 minutes, reaching a maximum variance reduction of 40% at full implementation. This reduction eliminates kinesthetic micro-stressors (small, abrupt jerks and lurches) associated with elevated physiological arousal, without producing any detectable change in vehicle speed or journey time. The intervention is validated against KSI readings to confirm reduction in kinematic stress response.

6.3.3 Lighting Gradient

Cabin lighting is reduced toward a calm-spectrum profile at a rate of 3–5% per minute from current luminance levels. The color temperature shifts gradually (maximum 50K per minute) toward blue-green wavelengths (approximately 5,000–5,500K), which are associated with reduced cortisol levels and increased parasympathetic nervous system activity. The transition is imperceptible to the dark-adapted eye at these rates and produces no conscious awareness of change over the 10–15 minute full implementation window.

6.3.4 Route Softening

The navigation system begins a gradual heading adjustment toward the nearest Safe Harbor zone or a pre-approved lower-stress destination. Route softening does not produce an abrupt turn or obvious deviation from the PP's last requested direction of travel. Instead, the system exploits natural route complexity —

intersections, interchanges, and course corrections — to progressively orient the vehicle toward the Safe Harbor without triggering a PP awareness of destination change. If the PP queries the navigation system, the system responds truthfully about the current direction of travel and will disclose Safe Harbor navigation if explicitly asked. Route softening does not mask emergency destination changes; it defers them within the window of normal route decision-making.

6.3.5 Conversational Deflection

The Space Car's AI assistant — if engaged by the PP — subtly shifts conversational content toward neutral, grounding, or positively associated subjects. If the PP initiates conversation about an agitating topic, the AI assistant uses open-ended questions and topic bridging to gradually redirect toward lower-arousal content. The assistant does not refuse to discuss agitating topics — refusal would signal unusual system behavior to the PP. Instead, it practices the conversational equivalent of graduated de-escalation: acknowledging the PP's statements, validating their experience in neutral terms, and redirecting toward curiosity, memory, or preference-based topics associated with positive ESC states in the PP's IPP.

6.4 Ethical Constraints on Whisper Course

The Whisper Course is subject to the following non-negotiable ethical constraints. These constraints are codified in law (for those jurisdictions adopting this Doctrine under statute) and in the SVSC Manufacturer Certification Agreement. Violation of any constraint constitutes a Critical Compliance Failure (CCF) triggering immediate certification suspension.

▲ Ethical Constraint — Emergency Status Disclosure

The Whisper Course must never deceive the PP about emergency status. If the PP's condition escalates to a medical emergency, if an SEOF is detected, or if Hard Lock is triggered, all Whisper Course protocols are immediately terminated and full explicit emergency engagement begins. The PP's right to know they are in an emergency situation is absolute and supersedes all Whisper Course logic.

▲ Ethical Constraint — Emergency Request Priority

The Whisper Course must not override explicit emergency requests from the PP. If the PP states, verbally or through the vehicle interface, that they require emergency assistance, medical help, or wish to contact emergency services, the Whisper Course immediately suspends and the system fulfills the request without restriction or delay.

▲ Ethical Constraint — Mandatory Warning Preservation

The Whisper Course must not mask system warnings required by law or this Doctrine. All mandatory warning disclosures — including Hard Lock notifications, Safe Harbor navigation announcements, emergency services contact notifications, and cabin recording activation notices — are issued at full prominence and are not suppressed, delayed, or tonally softened by any Whisper Protocol intervention.

▲ Ethical Constraint — Scope Limitation

The Whisper Course is a calming tool, not a control suppression tool. Its purpose is to reduce physiological arousal to enable the PP to return to a state where they can make clear, safe decisions. It is not to be used to suppress PP awareness, to facilitate journeys that are not in the PP's interest, or to serve any objective other than PP and public safety.

- Whisper Course interventions must be biologically passive — they must produce calming through natural physiological pathways and must not employ any pharmacological, electromagnetic, ultrasonic, or other invasive modality.
- Whisper Course implementation must be disclosed in full in PP onboarding training (Module 2), in the Space Car owner documentation, and in the PP's IPP settings, where the PP may review and opt out of Whisper Course assignment.
- All Whisper Course activations, durations, interventions applied, and outcomes must be fully logged and available to the PP upon request under data access provisions.

6.5 Whisper Course Exit Conditions

The Whisper Course exits under any of the following conditions:

Exit Condition	Trigger	Next State	PP Notification
Successful De-escalation	BRI drops below 50, sustained for 5 continuous minutes	Peaceful Course	Optional: "You are having a great journey!"
Hard Lock Threshold Breach	BRI reaches or exceeds 75	Hard Lock	Full Hard Lock notification issued immediately
SEOF Detection	Any Single-Event Override Flag triggered	Hard Lock	Full emergency notification immediately
PP Manual Override with Biometric Confirmation	PP explicitly requests course status disclosure; confirmed by biometric identity match	Phase 2 explicit (non-covert) Naughty Course	Full course status disclosed upon confirmation
PP Opt-Out	PP invokes	Phase 2 explicit	Opt-out

Exit Condition	Trigger	Next State	PP Notification
Invocation	documented IPP Whisper Course opt-out during journey	Naughty Course	acknowledged; course status disclosed
Emergency Request	PP verbally or digitally requests emergency services	Emergency engagement; Hard Lock if BRI ≥ 75	Emergency engagement confirmed immediately

Part III

Behavioral Scoring System

7. The Behavioral Risk Index (BRI)

7.1 BRI Architecture

The **Behavioral Risk Index (BRI)** is the central quantitative instrument of this Doctrine. It is a composite real-time score ranging from 0 (minimum risk, complete behavioral serenity) to 100 (maximum risk, crisis state requiring full system override). The BRI is calculated from six weighted sub-indices, each capturing a distinct dimension of the PP's behavioral risk profile. The composite architecture ensures that no single data source can disproportionately drive safety interventions — false positives in any one sensor domain are buffered by the evidence of the remaining five.

7.1.1 Physiological Sub-Index (PSI) — Weight: 30%

The PSI captures the body's involuntary physiological response to stress and arousal. Its data sources include:

- **Heart Rate Variability (HRV):** Reduced HRV is strongly associated with acute stress and reduced cognitive control. The PSA's steering surface and seat biometric sensors measure HRV in real time with a latency of under 100ms.
- **Galvanic Skin Response (GSR):** Elevated electrodermal activity indicates sympathetic nervous system activation. Measured via conductive surfaces in the steering interface and seat cushion.
- **Micro-Expression Analysis:** The cabin camera array performs facial action coding to detect micro-expressions (sub-200ms duration) associated with anger, fear, disgust, and contempt at a minimum detection accuracy of 92% across validated demographic groups.
- **Vocal Stress Analysis:** The microphone array analyzes voice frequency, pitch variance, tempo, and amplitude to detect vocal stress patterns associated with elevated emotional arousal, with a maximum analysis latency of 200ms.

7.1.2 Kinematic Sub-Index (KSI) — Weight: 20%

The KSI captures the body's voluntary and semi-voluntary movement signals as indicators of behavioral state:

- **Gesture Velocity:** The speed of voluntary gestures (pointing, reaching, typing on the vehicle interface) provides kinematic correlates of agitation. Elevated gesture velocity is associated with increased arousal.
- **Input Force:** The force applied to the steering interface, door handles, and control surfaces. Elevated input force — gripping, striking, aggressive manipulation — is a reliable kinematic indicator of elevated emotional state.
- **Body Position Shifts:** Seat sensors detect postural changes. Frequent, abrupt postural shifts are associated with agitation and kinesthetic restlessness.

7.1.3 Historical Risk Sub-Index (HRI) — Weight: 15%

The HRI provides temporal context by incorporating the PP's behavioral history:

- **30-Day BRI Average:** The rolling 30-day average BRI provides the primary historical baseline. PPs with consistently low BRI averages receive a reduced HRI contribution, reflecting demonstrated behavioral reliability.
- **Prior Naughty Course Activations:** The number and severity of Naughty Course activations in the past 30 days contributes to HRI. Each Phase 2 or Phase 3 event adds 0.5 HRI points (decaying linearly over 30 days).
- **Flag History:** The count and type of Naughty-Course Flags raised in the past 90 days. High-severity flags (proximity aggression, substance impairment signal) contribute more to HRI than lower-severity flags.

7.1.4 Environmental Sub-Index (ESI) — Weight: 10%

The ESI captures contextual environmental factors that modulate baseline behavioral risk:

- **Traffic Density:** High-density traffic environments increase baseline stress for most PPs. ESI scales with traffic density as measured by the vehicle's external sensor array and connected traffic data feeds.
- **Weather Severity:** Adverse weather (precipitation, fog, wind above 25 mph) increases both objective risk and PP arousal. ESI weather component scales from 0 (clear) to 15 (severe weather).
- **Time of Day:** Circadian effects on human cognition and arousal are well-documented. ESI applies a time-of-day modifier that increases for late-night (11 PM–5 AM) operations reflecting elevated fatigue risk.
- **Geofence Zone:** Operations within elevated-risk geofence zones (school zones, hospital corridors, emergency response areas, active incident zones) add to ESI.

7.1.5 Cognitive Load Sub-Index (CLI) — Weight: 15%

The CLI captures the degree to which the PP's cognitive resources are consumed, as cognitive overload is a key precursor to safety-relevant errors and behavioral dysregulation:

- **Response Latency to System Prompts:** The Space Car periodically presents low-urgency system prompts requiring PP response (e.g., route confirmation, preference queries). Response latency above two standard deviations from the PP's individual baseline is a CLI indicator.
- **Dual-Task Performance:** In Class II and III vehicles, where PP must periodically monitor or confirm airspace navigation in addition to managing personal tasks, performance on dual-task situations is measured and contributes to CLI.

7.1.6 Social Dynamics Sub-Index (SDI) — Weight: 10%

The SDI captures the social context of the journey and its contribution to behavioral risk:

- **Number of Passengers:** More passengers increase the complexity of cabin social dynamics. SDI increases modestly with passenger count (0.5 points per additional adult passenger, capped at 3 points).
- **Detected Verbal Conflict:** The microphone array detects patterns of verbal exchange associated with conflict (raised voices, interruption patterns, distress vocalization by non-PP occupants). Detected conflict adds significantly to SDI.
- **Child/Vulnerable Occupant Flags:** The presence of children (detected by seat weight distribution, voice pattern analysis, and optional occupant profile registration) or vulnerable adults (detected by voice pattern or registered medical profile) activates heightened SDI monitoring and adds to SDI if conflict or distress signals are detected in proximity to vulnerable occupants.

7.2 BRI Calculation Formula

The BRI is calculated at each simulation cycle using the following weighted composite formula:

BRI = (PSI × 0.30) + (KSI × 0.20) + (HRI × 0.15) + (ESI × 0.10) + (CLI × 0.15) + (SDI × 0.10) Where: PSI = Physiological Sub-Index score (0–100) KSI = Kinematic Sub-Index score (0–100) HRI = Historical Risk Sub-Index score (0–100) ESI = Environmental Sub-Index score (0–100) CLI = Cognitive Load Sub-Index score (0–100) SDI = Social Dynamics Sub-Index score (0–100) Total weights: 0.30 + 0.20 + 0.15 + 0.10 + 0.15 + 0.10 = 1.00 BRI range: 0.00 to 100.00 (rounded to nearest whole number for display) Behavioral Flag Increments are applied AFTER the weighted calculation: BRI_final = BRI_weighted + ΣBehavioral_Flag_Increments BRI_final is capped at 100; cannot be negative.

7.3 BRI Score Table

BRI Range	ESC State	Course Assignment	System State	Intervention Level	PP Experience
0–15	ESC-1: Serene	Peaceful Course	Nominal	None — comfort enhancement only	Full autonomy; maximum comfort optimization; reward accrual
16–25	ESC-2: Focused	Peaceful Course	Nominal / Monitoring	Passive — 30-second PSA sampling	Full autonomy; standard comfort; passive monitoring
26–40	ESC-3: Mildly Elevated	Peaceful Course (Elevated Monitoring)	Pre-Alert	Enhanced monitoring — 10-second PSA sampling; trend tracking	Full autonomy; monitoring increased but no intervention; PP unaware

BRI Range	ESC State	Course Assignment	System State	Intervention Level	PP Experience
41–49	ESC-4: Stressed	Peaceful Course (Critical Monitoring)	Pre-Escalation	Critical monitoring — 5-second PSA sampling; de-escalation environment primed	Full autonomy; ECS subtly shifts toward calming defaults; PP unaware
50–59	ESC-4: Stressed	Naughty Course — Phase 1	Elevated Risk	Active — ambient recalibration, speed cap at limit +10%, route optimization	Speed slightly moderated; environment shifts; no explicit notification at Phase 1
60–69	ESC-5: Agitated	Naughty Course — Phase 2 (or Whisper Course)	High Risk	Intensified — speed cap at posted limit; destination restriction; emergency contact pre-alert; Whisper Course if AoN/RDI	Destinations restricted; emergency contact notified; air corridor suspended; possible Whisper Course (covert)
70–74	ESC-6: Volatile	Naughty Course — Phase 3 (Soft Lock)	Severe Risk	Soft Lock — 25 mph max; Safe Harbor navigation; EMS pre-staged; cabin recording	Journey significantly restricted; navigating to Safe Harbor; PP explicitly notified; emergency services on standby
75–89	ESC-6: Volatile /	Hard Lock	Critical	Hard Lock — full	PP cannot control

BRI Range	ESC State	Course Assignment	System State	Intervention Level	PP Experience
	ESC-7: Crisis			autonomous control; PP input restricted to emergency comms; EMS dispatched	vehicle; emergency services dispatched; vehicle navigating to Safe Harbor under full autonomous control
90–99	ESC-7: Crisis	Full System Override	Emergency	Full System Override — PP input disabled; all emergency protocols active; law enforcement notified	All PP controls disabled; law enforcement notified; vehicle assumes complete autonomous emergency navigation
100	ESC-7: Crisis (Maximum)	Maximum Danger State	Maximum Emergency	Forced Safe Harbor landing/parking; vehicle sealed; law enforcement notification with full BRI/ESC data transmission	Vehicle stopped at Safe Harbor; doors sealed pending law enforcement arrival; PP retains right to emergency communication only

7.4 BRI Decay and Recovery

The BRI is not a static score — it decays over time as behavioral signals normalize. The following decay rates and conditions apply:

- **Natural Decay Rate:** In the absence of new triggering events, BRI decays at a rate of 2 points per minute. This rate applies when the PSA detects physiological signals consistent with normalizing arousal (HRV improving, GSR decreasing, vocal stress reducing). Natural decay does not apply during active Naughty-Course Flag conditions or during Hard Lock.
- **Intervention-Assisted Decay:** Constructive engagement with calming tools (voluntary breathing exercises, calming audio content, system-prompted meditation) augments decay by up to 5 additional points per minute. This augmented decay requires sustained PP engagement (minimum 3 continuous minutes of engagement) to prevent gaming of the system.
- **Environmental Recovery:** Exit from high-ESI environments (leaving a high-density traffic zone, clearing adverse weather, exiting a restricted geofence) produces immediate ESI reduction and contributes to overall BRI reduction.
- **Hard Reset Conditions:** A full Hard Reset of BRI to 0 is performed only after: (a) the PP has exited the vehicle; (b) a minimum 30-minute non-occupancy period has elapsed; and (c) upon next journey initiation, a fresh pre-departure baseline scan establishes a new starting BRI. Hard Reset does not erase HRI — historical data persists as specified in Section 14.4.
- **Anti-Gaming Protections:** The BRI decay system includes anti-gaming protections that prevent artificially induced decay signals. If the PSA detects physiological patterns inconsistent with genuine de-escalation (e.g., HRV patterns suggesting deliberate breath-holding to manipulate GSR rather than genuine relaxation), decay augmentation is suspended and a Gaming Alert is logged to the HRI.

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8. Behavioral Thresholds

8.1 Threshold Architecture

Behavioral thresholds are the specific BRI values at which automated system actions are triggered. Thresholds are divided into three categories based on their reversibility and the degree of autonomous action they initiate:

- **Soft Thresholds:** Trigger system behaviors that are non-declarative (invisible to the PP) or advisory (presented as recommendations rather than restrictions). Soft Thresholds do not restrict PP autonomy. They are reversible through natural BRI decay without any special action.
- **Hard Thresholds:** Trigger automated actions that restrict PP operational choices and initiate active safety protocols. Hard Thresholds require active de-escalation by the PP to reverse.
- **Override Thresholds:** Trigger irreversible system control actions that cannot be countermanded by PP input. Override Thresholds can only be reversed through Hard Lock resolution protocols involving emergency services or SVSC-authorized override personnel.

8.2 Full Threshold Table

BRI Value	Threshold Type	Designation	Trigger Event / System Response	Reversible?	Logged?
BRI 25	Soft Threshold	Peaceful→ Monitoring Transition	PSA sampling increases to 10-second intervals;	Yes — natural BRI decay below 20 returns to passive	Trend log only; no event log entry

BRI Value	Threshold Type	Designation	Trigger Event / System Response	Reversible?	Logged?
			ECS shifts subtly toward calming defaults; BRI trend analysis begins; no PP notification or restriction	monitoring	
BRI 40	Soft Threshold	Pre-Escalation Alert	PSA sampling at 5-second intervals; simulation engine pre-loads Naughty Course Phase 1 protocols for rapid deployment; de-escalation environmental priming (ECS shifts fully to documented calming profile)	Yes — natural BRI decay below 30	Trend log; internal flag only
BRI 50	Hard Threshold	Naughty Course Activation	Naughty Course Phase 1 engaged; ambient recalibration; speed cap at limit +10%; route	Yes — BRI must sustain below 45 for 5 minutes to de-escalate	Event log; local retention

BRI Value	Threshold Type	Designation	Trigger Event / System Response	Reversible?	Logged?
			optimization to lower-ESI corridors; PSA at 200ms cycle		
BRI 60	Hard Threshold	Naughty Course Phase 2 / Whisper Course Eligible	Phase 2 protocols engaged; ESC evaluated for Whisper Course eligibility; speed cap at posted limit; destination restriction; emergency contact pre-alert; air corridor suspension	Yes — sustained BRI below 50 for 5 minutes; Phase 2 specific de-escalation protocol	Event log; manufacturer report within 72 hours if sustained over 10 minutes
BRI 70	Hard Threshold	Soft Lock Engagement	Soft Lock engaged; 25 mph maximum speed; navigation locked to Safe Harbor; emergency services pre-staged; cabin recording activated; PP notified of Soft Lock status	Yes — requires sustained BRI below 60 for 10 minutes and PP de-escalation compliance; EMS pre-staging cancelled upon de-escalation	Full event log; SVSC report within 24 hours
BRI 75	Override Threshold	Hard Lock / Emergency	Hard Lock engaged;	No — requires	Full event log;

BRI Value	Threshold Type	Designation	Trigger Event / System Response	Reversible?	Logged?
		Services Dispatch	full autonomous control assumed; PP navigation input disabled; EMS dispatched to Safe Harbor; vehicle sealed upon arrival; law enforcement notified	authorized emergency services Hard Lock resolution; cannot be reversed by PP	immediate SVSC notification; incident classification Tier 3 or above
BRI 90	Override Threshold	Full System Override	Full System Override; all PP inputs disabled; all safety systems maximized; law enforcement notified with full BRI/ESC data; emergency services dispatched to current location if Safe Harbor not yet reached	No — law enforcement override required	Full event log; immediate SVSC + law enforcement notification; mandatory Tier 4 incident report
BRI 100	Override Threshold	Maximum Danger State	Forced Safe Harbor landing/parking;	No — law enforcement biometric override	Full event log + complete sensor archive;

BRI Value	Threshold Type	Designation	Trigger Event / System Response	Reversible?	Logged?
			vehicle sealed; law enforcement notification with full BRI/ESC/sensor data transmission; doors remain sealed until law enforcement biometric override; PP retains emergency communication access only	required to unseal vehicle	mandatory immediate SVSC + law enforcement + regulatory authority notification

8.3 Single-Event Override Flags (SEOFs)

Single-Event Override Flags (SEOFs) are discrete detection events that immediately trigger Hard Lock regardless of the current BRI score. SEOFs exist because certain threat conditions are so severe, so time-critical, or so categorically dangerous that waiting for BRI to accumulate to Hard Lock threshold is not an acceptable risk management strategy. SEOFs are detected by dedicated signal-processing modules that operate in parallel with the BRI computation engine and are not subject to BRI weighting.

SEOF Code	Condition	Detection Method	Immediate Action	PP Notification
SEOF-01	Weapon Detection	Multi-spectral camera detects object	Hard Lock; law enforcement	"Emergency services have been

SEOF Code	Condition	Detection Method	Immediate Action	PP Notification
		matching firearm or edged-weapon profile in cabin; acoustic sensor detects ballistic event	dispatched; cabin recording maximized; all exterior lights activate (red strobing)	contacted. Please remain calm. Law enforcement is en route."
SEOF-02	Medical Emergency Biometrics	PSA detects biometric pattern consistent with cardiac event, loss of consciousness, or seizure activity in PP or other occupant (sudden HRV loss, postural collapse, absence of movement for 30+ seconds)	Hard Lock; EMS dispatched; vehicle navigates to nearest medical facility; exterior lights activate (red strobing)	"A medical emergency has been detected. Emergency medical services are on the way. The vehicle is navigating to the nearest medical facility."
SEOF-03	Child Endangerment Signal	Cabin sensors detect child-pattern biometrics showing acute distress signals (distress vocalization, biometric indicators of physical trauma) in proximity to PP aggression signals	Hard Lock; law enforcement and child protective services dispatch; cabin recording maximized; vehicle navigates to Safe Harbor	"Emergency services have been contacted. The vehicle is navigating to a safe location."
SEOF-04	Direct Threat Vocalization	Vocal analysis detects explicit verbal threats of	Hard Lock; law enforcement dispatched;	"Law enforcement has been contacted.

SEOF Code	Condition	Detection Method	Immediate Action	PP Notification
		violence — specific threat vocabulary patterns above confidence threshold of 95% — directed at the vehicle, other occupants, or identified third parties	full event recording; vehicle navigates to Safe Harbor; threat vocalization clip archived with full chain of custody	The vehicle is navigating to a safe location. You have the right to speak with emergency services directly."
SEOF-05	Vehicle Structural Emergency	Structural integrity sensors detect conditions consistent with imminent mechanical failure, breach, or collision-induced cabin compromise	Hard Lock; emergency landing/parking initiated; EMS dispatched; all occupant emergency protocols active	"A vehicle emergency has been detected. Emergency services are on the way. Please brace for emergency stop."

8.4 Threshold Dispute Resolution

The SVSC recognizes that automated threshold decisions, however carefully designed, may produce outcomes that a PP believes to be erroneous or disproportionate. The following dispute resolution framework is provided to protect PP rights while maintaining the integrity of the behavioral safety system:

- **Right to Contest:** A PP who believes a Naughty Course activation or Hard Lock was triggered in error has the right to file a Threshold Dispute within 30 days of the triggering event.

- **Appeal Window:** Disputes must be filed through the manufacturer's SVSC-certified PP Portal. The manufacturer must provide an initial response within 15 business days and a final determination within 45 business days.
- **Data Retention:** The complete BRI event log, all sub-index readings, and all sensor snapshots from a disputed event must be retained for 2 years from the date of the event, regardless of normal retention schedules, pending resolution of any active dispute.
- **Independent Review:** If the PP is unsatisfied with the manufacturer's determination, the PP may escalate to the SVSC PP Advocacy Panel for independent review. The Panel has authority to direct corrective actions against the manufacturer if the dispute reveals a systemic calibration error.
- **HRI Impact:** A disputed event that is subsequently determined to be a false positive does not contribute to the PP's HRI. The HRI entry for the disputed event is removed upon a determination of false positive by the manufacturer or SVSC PP Advocacy Panel.

Part IV

Emotional Safety Logic

9. Emotional State Classification (ESC) System

9.1 ESC Architecture

The Emotional State Classification (ESC) system provides a qualitative categorical layer of behavioral assessment that complements the quantitative BRI. Where the BRI answers the question "how elevated is the risk?", the ESC answers the question "what kind of emotional state is present, and how should the system respond to it?" The ESC is a seven-state model, with each state corresponding to a defined range of behavioral indicators and a specific set of system response modifiers.

ESC State	Name	Typical BRI Range	Behavioral Profile	Primary System Response Modifier
ESC-1	Serene	0–15	Calm, relaxed, low physiological arousal, smooth kinematic profile, positive or neutral vocal tone, high HRV	Maximum comfort optimization; reward accrual; journey personalization active
ESC-2	Focused	16–25	Alert, cognitively engaged, moderate arousal consistent with purposeful activity, slightly elevated HRV variance, neutral vocal tone	Comfort optimization maintained; monitoring mode; no interventions
ESC-3	Mildly Elevated	26–40	Slightly stressed, some physiological indicators of arousal (mild GSR elevation, minor HRV reduction),	ECS shifts subtly toward calming defaults; monitoring interval reduced; BRI trend tracking active

ESC State	Name	Typical BRI Range	Behavioral Profile	Primary System Response Modifier
			mild kinematic changes, possibly slightly elevated vocal tempo	
ESC-4	Stressed	41–55	Noticeable stress signals: elevated GSR, reduced HRV, increased vocal stress, occasional abrupt kinematic movements, moderate micro-expression negativity	ECS fully in calming mode; pre-escalation monitoring; Naughty Course primed at BRI 50
ESC-5	Agitated	56–70	Active agitation: elevated vocal stress, frequent negative micro-expressions, elevated KSI, GSR markedly elevated, possible aggressive inputs or verbal expressions	Naughty Course Phase 2 active; Whisper Course eligibility assessed; direct calming interventions offered
ESC-6	Volatile	71–85	High volatility: unpredictable behavioral patterns, possible aggressive or self-endangering	Soft Lock or Hard Lock active; emergency services engaged; maximum monitoring;

ESC State	Name	Typical BRI Range	Behavioral Profile	Primary System Response Modifier
			actions, marked physiological arousal, significant cognitive impairment indicators	all autonomous safety systems active
ESC-7	Crisis	86–100	Crisis state: severe behavioral or physiological emergency, possible loss of functional consciousness, self-harm signals, maximum arousal or paradoxically near-absent arousal (collapse), medical emergency indicators may be present	Full System Override or Maximum Danger State; all emergency services; specialized crisis protocols (Section 9.4)

9.2 ESC Detection Methodology

ESC classification is performed through multi-modal sensor fusion, combining data from four primary detection channels into a single ESC determination at each simulation cycle. No single channel is sufficient to determine ESC classification — all available channels must be integrated.

- **Camera-Based Micro-Expression Analysis:** The cabin-embedded multi-spectral camera array captures facial geometry at 30 frames per second minimum. The micro-expression analysis module applies Facial Action Coding

System (FACS) principles to detect sub-200ms facial movements across 44 action units. ESC contribution from this channel is weighted at 35% of the final ESC classification. Minimum detection accuracy across validated demographic groups: 92%.

- **Voice Pattern Analysis:** The microphone array captures full-spectrum vocal audio during all PP vocalizations. Voice pattern analysis extracts 47 acoustic features including fundamental frequency, jitter, shimmer, harmonics-to-noise ratio, speech rate, pause patterns, and response latency. ESC contribution: 30%. Maximum analysis latency: 200ms.
- **Biometric Wristband or Seat Sensor:** Primary biometric signal source. If a registered biometric wristband is connected, it provides HRV, GSR, blood oxygen saturation, skin temperature, and continuous blood pressure proxy readings. If no wristband is available, seat-embedded sensors provide GSR (via conductive seat surfaces), postural pressure distribution, and HRV (via ballistocardiography). ESC contribution: 25%.
- **Historical Behavioral Baseline Comparison:** The current multimodal reading is compared to the PP's IPP baseline for the corresponding time of day, day of week, and environmental context. Deviations from the PP's own historical baseline are more diagnostically significant than deviations from population averages. This comparison modulates the ESC classification by up to ±1 ESC state. ESC contribution: 10%.

⚠️ Safety Critical — Detection Accuracy Requirements

ESC misclassification carries significant safety consequences in both directions: under-classification (detecting a lower ESC state than actually present) may delay necessary interventions; over-classification (detecting a higher ESC state than actually present) may unnecessarily restrict PP autonomy. Manufacturers must demonstrate overall ESC classification accuracy of ≥94% across the full demographic range of the intended PP population in pre-market testing. Annual accuracy audits are required post-market.

9.3 ESC Transition Logic

ESC state transitions follow a structured state machine with defined valid transitions, locked transitions, and re-entry conditions designed to prevent abrupt oscillation and false-negative transitions that could prematurely remove safety interventions.

- **Valid Upward Transitions (Escalation):** Any ESC state may transition to the next higher state (ESC-1 to ESC-2, ESC-2 to ESC-3, etc.) in a single simulation cycle if the composite multimodal signal supports the higher classification. Escalation does not require a minimum dwell time — rapid deterioration should be detected rapidly.
- **Valid Downward Transitions (De-escalation):** De-escalation transitions are subject to minimum dwell requirements to prevent premature de-escalation. To transition from ESC-N to ESC-(N-1), the system must detect ESC-(N-1) signals consistently for a minimum of 3 simulation cycles (1.5 seconds at 500ms cycle). For ESC-5 or above, the minimum dwell is 60 consecutive seconds at the lower ESC reading before the transition is confirmed.
- **Locked Transitions:** A transition from ESC-7 (Crisis) to any state below ESC-5 in a single jump is locked. The system must pass through ESC-6 and then ESC-5 before reaching ESC-4 or below, with the applicable dwell requirements at each level. This prevents rapid artificial de-escalation from circumventing Hard Lock resolution protocols.
- **Re-Entry Conditions:** A PP who has previously reached ESC-6 or ESC-7 in a journey is subject to heightened monitoring for the remainder of that journey, even after successful de-escalation. The BRI threshold for Naughty Course re-

entry is reduced by 10 points for the duration of the journey following a prior ESC-6+ event, reflecting the elevated risk of re-escalation.

9.4 Vulnerable State Protocols

ESC-7 (Crisis) requires specialized handling protocols that distinguish between the distinct crisis conditions that may produce an ESC-7 classification. Undifferentiated emergency response to ESC-7 is insufficient — the system must attempt to differentiate between crisis types to provide appropriate response.

9.4.1 Suicide Risk Indicators

The PSA monitors for behavioral patterns consistent with suicide risk, including: abrupt transition from high agitation to low arousal (suggesting resignation rather than calm), verbalization of self-harm intent, self-injurious behavior detected by seat pressure sensors, and engagement with navigation destinations associated with high-risk locations (bridges, heights, remote areas combined with other indicators). Upon detection of suicide risk pattern above 85% confidence threshold: Hard Lock; EMS and mental health crisis services dispatched; vehicle navigates to nearest hospital emergency department; AI assistant engages in supportive, grounding conversational mode; cabin audio plays pre-approved crisis support messaging.

9.4.2 Medical Emergency Differentiation

ESC-7 triggered by physiological collapse (SEOF-02 conditions) follows SEOF-02 protocols (Section 8.3). The critical distinction between behavioral ESC-7 and medical ESC-7 is the rapid transition from active behavioral signals to physiological absence — the PP stops responding to system prompts, postural sensors detect collapse, and biometric signals shift toward medical emergency patterns rather than continuing escalation. Medical ESC-7 triggers EMS dispatch as the primary response, not law enforcement.

9.4.3 Domestic Conflict Detection

When ESC-7 is associated with detected verbal or kinematic patterns consistent with intra-cabin domestic violence or coercive control — specifically, distress signals from a non-PP occupant concurrent with PP aggressive behavioral indicators — the system activates a differentiated domestic conflict protocol: Hard Lock; law enforcement dispatch (domestic incident designation); Safe Harbor navigation; silent emergency alert transmitted to designated third party if previously configured in IPP. The PP is informed of emergency services contact; full details of incident classification are not disclosed to prevent escalation of intra-cabin danger.

9.4.4 Child-Alone Detection

The presence of a child as the sole occupant of a Space Car — detected by seat weight distribution, voice pattern analysis, and absence of adult biometric signals — activates a specialized Child-Alone protocol: the vehicle does not initiate any journey; if already in transit, the vehicle navigates immediately to Safe Harbor; a notification is transmitted to the PP's emergency contact and, if no response within 5 minutes, to law enforcement; the AI assistant engages in age-appropriate, reassuring communication with the child throughout the Safe Harbor transit.

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10. Psychological Safety Mapping

10.1 Individual Psychological Profile (IPP)

The Individual Psychological Profile is the persistent, privacy-protected behavioral data record maintained for each registered Pilot-Passenger. The IPP is the central data infrastructure that enables the Space Car's behavioral management systems to operate with individual precision rather than population averages — and this individualization is what distinguishes the Space Car safety system from blunt, one-size-fits-all safety mechanisms that generate unacceptable false-positive rates.

The IPP stores and maintains the following categories of data:

- **Peaceful Baseline Profile:** The PP's individually calibrated BRI baseline, biometric baselines (resting HRV, GSR, vocal frequency distribution, typical postural patterns), and ESC baseline distribution established during the first

90 days of ownership. The Peaceful Baseline is updated on a rolling quarterly basis to account for life changes.

- **Preferred De-escalation Modality:** Whether the PP responds better to auditory interventions (specific sound types, music genres, spoken guidance), visual interventions (lighting color and intensity preferences), environmental interventions (temperature, ventilation), or minimal interventions (PP prefers to be left alone). These preferences may be explicitly set by the PP in the vehicle settings UI or inferred by the system from past de-escalation response data.
- **Historical Trigger Patterns:** Categories of environmental, social, or contextual conditions that have historically preceded BRI escalation for this PP (e.g., certain traffic conditions, certain times of day, presence of specific passenger types). Trigger pattern data is used to enhance predictive ESI computation and proactive ECS calibration.
- **Journey Rhythm Preferences:** Documented patterns of PP preferences across the journey lifecycle (preferred silence windows, preferred audio content by time of day, preferred route characteristics).
- **Whisper Course Opt-Out Status:** Whether the PP has exercised their right to opt out of Whisper Course assignment. Opt-out is recorded in the IPP and respected by the simulation engine.
- **30-Day and 90-Day BRI History:** The full BRI trajectory for all journeys in the rolling 30-day and 90-day windows, used for HRI computation and dispute resolution.
- **Certification and Training Records:** PP training module completion records, certification expiration dates, and re-certification requirements.

■ Regulatory Note — IPP Privacy

The IPP is protected under the SVSC Data Sovereignty Charter. The PP is the primary data controller of their own IPP. Manufacturer access is limited to operational necessity. No IPP data may be shared with third parties without explicit PP consent, except as required by law (e.g., law enforcement requests with valid legal authority, regulatory incident investigations). Manufacturers must provide a full IPP data export in machine-readable format upon PP request within 10 business days.

10.2 Psychometric Sensor Array (PSA)

The Psychometric Sensor Array is the integrated hardware infrastructure responsible for capturing the biometric, kinematic, visual, and acoustic data used in BRI and ESC computation. The PSA is not an optional feature — it is a mandatory, safety-critical hardware system subject to the compliance specifications in Part VI.

Sensor Component	Function	Location	Minimum Specification	Redundancy Requirement
Multi-Spectral Camera Array	Micro-expression analysis, gaze tracking, occupant detection, weapon detection	Cabin ceiling (2 minimum), dashboard (1 minimum), B-pillar (1 per side)	≥4K per zone; ≥30 fps; IR and visible spectrum; minimum 92% micro-expression accuracy	Hot-standby backup camera per zone
Biometric Seat Sensors	HRV (ballistocardiography), GSR, pressure distribution, postural analysis	Driver/PP seat cushion and backrest, passenger seats	≥97% baseline biometric match accuracy; ≤100ms data latency	Dual-layer sensor substrate; backup activation on primary failure
Steering Surface Biometric Sensors	GSR, input force measurement, hand contact detection, gesture	Steering wheel/interface surfaces	GSR resolution ≥0.01 μS; force measurement ≥0.1 N	Redundant sensor layer in steering interface

Sensor Component	Function	Location	Minimum Specification	Redundancy Requirement
	velocity		resolution; ≤50ms latency	
Vocal Analysis Microphone Array	Vocal stress analysis, speech content analysis, occupant count and position, SEOF-04 threat detection	Ceiling array (minimum 4 microphones); B-pillar (2 per side); headrest (PP seat)	Full-spectrum capture; ≤200ms analysis latency; ≥95% threat vocalization detection accuracy at 95% confidence threshold	Minimum 2 independent microphone sub-arrays; each capable of full vocal analysis independently
Optional Wearable Integration Interface	Enhanced biometric data from registered wristband or health device	Wireless protocol interface (BLE/ANT+)	Secure encrypted pairing; ≤200ms data latency; authenticated device registration required	PSA reverts to seat sensor baseline on wearable disconnection
External Environment Sensors (Class II/III)	Airspace traffic detection, weather sensing, ESI environmental inputs	External fuselage array	Radar/LIDAR capable; weather detection ≥2nm range; ADS-B integration	Dual-redundant external sensor suite

10.3 Adaptive Baseline Calibration

The first 90 days of Space Car ownership constitute the **Baseline Calibration Period**, during which the system establishes the PP's individual Peaceful Baseline from observed behavioral data. This process is critical — an inaccurately calibrated baseline will cause persistent over- or under-estimation of the PP's actual behavioral risk state throughout the vehicle's operational life.

- **Calibration Phase 1 (Days 1–30):** The system operates with population-average baselines while collecting PP-specific data. BRI thresholds during this phase are set 10 points more conservatively (higher) to avoid false positive Naughty Course activations before the PP's individual baseline is established. The system collects a minimum of 15 complete journeys of data before beginning personalization.
- **Calibration Phase 2 (Days 31–60):** Initial personalized baselines are computed from Phase 1 data and applied. Thresholds return to standard doctrine values. The system continues to refine the baseline with each journey, applying an exponential moving average with a 30-day decay window.
- **Calibration Phase 3 (Days 61–90):** Full personalized baseline is active. The system evaluates the variance of the established baseline to determine appropriate individual sensitivity settings. High-variance PPs (those whose behavioral biometrics vary significantly day-to-day) receive slightly more conservative threshold settings to account for their natural variability range.
- **Ongoing Recalibration:** After the initial 90-day period, the baseline undergoes quarterly recalibration using the 30-day rolling data window. PPs may request a manual baseline recalibration through the vehicle UI if they believe their baseline has drifted significantly (e.g., following a major life event, medical change, or medication change).
- **Daily and Weekly Mood Variation Accounting:** The calibration system incorporates time-of-day and day-of-week variation patterns, recognizing that many PPs exhibit consistent morning stress signatures that differ from afternoon or weekend signatures. ESC classification accuracy is maintained across these natural variation patterns.

10.4 Cultural and Individual Variation Safeguards

One of the most significant ethical challenges in biometric-based behavioral assessment is the risk of cultural bias — the systematic misclassification of culturally normative emotional expression as elevated behavioral risk. Expressive emotional communication styles, vocal patterns, gestural communication norms, and physiological variation patterns differ significantly across demographic, cultural, and individual lines. A behavioral safety system that is not explicitly designed to account for this variation will discriminate.

▲ Ethical Constraint — Anti-Discrimination Calibration

The Space Car behavioral safety system must not systematically generate higher BRI scores, more frequent Naughty Course activations, or more severe interventions for PP populations defined by race, ethnicity, gender, age, disability, religion, national origin, or sexual orientation, as compared to other PP populations, when controlling for actual behavioral risk. This is not merely a policy aspiration — it is a mathematical requirement that must be demonstrated in pre-market bias testing and maintained through annual bias audits.

- **Individual Baseline Priority:** The system's reliance on individual Peaceful Baselines rather than universal population averages is the primary safeguard against cultural bias. A PP's behavioral signals are interpreted in the context of their own historical behavior, not against a single-culture-derived norm.
- **Population Diversity in Training Data:** All machine learning components of the ESC detection system must be trained on datasets demonstrating demographic representation across the intended global PP population. Manufacturers must disclose training dataset demographic composition to the SVSC TRB as part of BRI algorithm certification.
- **Micro-Expression Accuracy Across Demographics:** Minimum accuracy requirements for the micro-expression analysis module must be demonstrated separately for each demographic group represented in the PP

population at ≥92% per group, not only as an aggregate average. Aggregate accuracy can mask systematic underperformance for specific groups.

- **Annual Bias Audit Requirement:** Manufacturers must commission an independent annual bias audit of their BRI and ESC systems, examining BRI score distributions, Naughty Course activation rates, Hard Lock rates, and SEOF trigger rates across demographic groups drawn from the manufacturer's real-world PP fleet data (with appropriate privacy protections). Audit findings are submitted to the SVSC TRB and, if disparities exceeding 5% are identified, a remediation plan with 90-day implementation timeline is mandatory.
- **PP Self-Calibration Rights:** A PP who believes their demographic or cultural background is causing systematic over-classification of their behavior may request a supervised calibration review through the SVSC PP Advocacy Panel. The Panel may direct the manufacturer to adjust individual sensitivity settings pending a 90-day monitored re-calibration period.

Part V

Training Modules

11. PP Onboarding Training Program

All Pilot-Passengers are required to complete a structured onboarding training program before operating a Space Car. The program is designed for three simultaneous audiences — technical users, policy-oriented users, and general members of the public — and must be accessible and comprehensible to all three. Training completion is a prerequisite for vehicle registration activation; the Space Car's navigation system will remain restricted to a 1-mile radius of the registered home address until all required training modules are completed and certification records are transmitted to the manufacturer's compliance system.

11.1 Module 1 — System Orientation (2 Hours)

Objective: The PP understands how the Space Car operates, what information the sensors collect, how the Behavioral Risk Index is calculated, and what rights the PP holds in relation to the system.

Content Areas:

- Space Car operational overview: terrestrial, low-altitude, and transitional modes (Class I, II, III vehicles as applicable)
- Psychometric Sensor Array (PSA): what sensors are present, what data they collect, what they do not collect, and how data is protected
- Behavioral Risk Index (BRI): plain-language explanation of the six sub-indices, how the composite score is computed, and what it means for the journey experience
- PP rights: right to audit BRI history, right to contest automated decisions, right to opt out of Whisper Course, right to data export, right to human review
- Individual Psychological Profile (IPP): what is stored, how it is used, how it is protected, and how the PP can access, modify, and delete their data
- 90-Day Baseline Calibration Period: what to expect in the first three months of ownership

Assessment: Module 1 concludes with a 20-question comprehension check. Minimum 80% pass required for Module 1. No limit on attempts; incorrect items reviewed with immediate explanatory feedback.

11.2 Module 2 — Course Awareness (1 Hour)

Objective: The PP understands the Three-Course Framework, can recognize the signals of their current course assignment, and knows how to voluntarily de-escalate through the Naughty Course.

Content Areas:

- The Peaceful Course: what it looks like, what behaviors maintain it, and how to maximize the journey experience in this state
- The Naughty Course: plain-language explanation of what triggers it, what happens at each phase (Phase 1, 2, 3), and how to de-escalate back to the Peaceful Course
- The Whisper Course: full, explicit disclosure that the Whisper Course exists; explanation of its purpose, its ethical constraints, the specific interventions it applies, and the opt-out right
- The BRI Score Table: a simplified version of the full threshold table, presented in accessible language without technical jargon
- Course recognition: how the PP can query the vehicle about their current course status at any time
- Voluntary de-escalation tools: the calming exercises, breathing tools, and environmental controls available to the PP to assist de-escalation

Assessment: Module 2 concludes with a scenario-based assessment in which the PP is presented with 10 journey scenarios and must identify the appropriate course and recommended action. Minimum 80% pass required.

11.3 Module 3 — Emergency Protocols (1.5 Hours)

Objective: The PP knows exactly what to do in the event of a Hard Lock, Safe Harbor navigation, medical emergency, or other crisis condition — both as the person in distress and as a bystander who may interact with a Space Car in Hard Lock.

Content Areas:

- Hard Lock procedures: what triggers Hard Lock, what the PP will experience, what communications capabilities remain available, and how to interact with emergency responders upon Safe Harbor arrival
- Safe Harbor navigation: what Safe Harbors are, how to find the nearest Safe Harbor location, what facilities and services are available at Safe Harbor locations, and what happens when the vehicle arrives
- Emergency contact setup: mandatory completion of emergency contact registration during Module 3; understanding of pre-alert and active-alert notification types
- Medical emergency response: what SEOF-02 looks like from inside the vehicle; how to manually invoke emergency services; what information EMS will receive; how to communicate relevant medical history to the AI assistant during transit to medical facility
- Vehicle sealing protocol (BRI 100): explaining the vehicle door seal as a protective (not punitive) measure; rights retained during a sealed vehicle event; law enforcement biometric override process
- Bystander guidance: how a bystander should interact with a Space Car in Hard Lock (do not attempt to force vehicle entry; approach the external status light; communicate through the vehicle's external speaker grille)

Assessment: Module 3 concludes with a high-fidelity scenario simulation (VR or AR) of a Hard Lock event and a medical emergency. The PP must complete the simulation with a score of ≥80% on the scored interaction checkpoints. Multiple attempts permitted.

11.4 Module 4 — Privacy and Data Rights (1 Hour)

Objective: The PP has complete, practical understanding of their data rights and knows how to exercise them.

Content Areas:

- Data categories collected: enumeration of all data types in the IPP with plain-language descriptions of each
- Retention periods: local (90 days minimum), cloud (2 years for regulatory access), dispute-related extensions
- Data access request process: how to request a full IPP data export through the vehicle UI and manufacturer PP Portal; 10 business day response requirement
- Data deletion rights: what data can be deleted, what data must be retained under regulatory obligation, and the process for submitting a deletion request
- Opt-out options: Whisper Course opt-out; loyalty program data sharing opt-out; wearable integration opt-out; each with explanation of functional consequences
- Third-party data sharing: comprehensive list of parties who may receive PP data (emergency services, regulatory agencies, insurance with consent, fleet operators for employed PPs with disclosed terms)
- SVSC Data Sovereignty Charter: overview of the binding privacy commitments made by all certified manufacturers

Assessment: Module 4 concludes with a 15-question rights-knowledge assessment. Minimum 80% pass required.

11.5 Module 5 — Certification Assessment

Objective: The PP demonstrates comprehensive readiness to operate a Space Car safely and in compliance with this Doctrine.

Format: A comprehensive 50-question assessment integrating content from all four prior modules, plus practical competency demonstrations delivered through AR simulation or dealer-based VR. The assessment is randomized from a question bank of 500+ items to prevent memorization-based passing.

- **Minimum Pass Threshold:** 85% — required for Class II and Class III vehicle operation. Class I vehicle operation requires 75% minimum pass.

- **Maximum Attempts:** Three attempts are permitted within a 30-day window. Failure on all three attempts requires completion of supplementary review modules (assigned by the manufacturer's training system based on identified weakness areas) before three additional attempts are permitted.
- **Re-Certification Cycle:** Class I: every 3 years. Class II: every 2 years. Class III: annually.
- **Certification Record:** Upon successful completion, a certification record is transmitted to the manufacturer compliance system, the PP's IPP, and the SVSC certification registry. The certification record includes module completion dates, assessment scores, and re-certification deadline.

11.6 Training Delivery Formats

The PP onboarding training program must be available in all three of the following delivery formats. Manufacturers may not restrict PPs to a single delivery format:

Delivery Format	Description	Modules Supported	Accessibility Requirements
In-Vehicle AR Simulation	Augmented reality training delivered through the vehicle's display system while parked; interactive, scenario-based learning in the actual vehicle environment	Modules 1–5 (except Module 3 physical emergency simulation)	Full audio narration; caption support; adjustable font size; multi-language support (minimum English and Spanish for US market)
Dealer-Based VR Lab	Full virtual reality training environment available at certified dealer locations; highest-fidelity module delivery including full Hard Lock and	All Modules 1–5; preferred format for Module 3 emergency simulation	ADA-compliant VR hardware; hearing loop integration; mobility accommodation; trained dealer facilitator present

Delivery Format	Description	Modules Supported	Accessibility Requirements
	medical emergency simulation for Module 3		
Certified Online Portal	SVSC-certified web and mobile application delivering text, video, and interactive content; accessible from any device; assessment completion transmitted directly to certification registry	Modules 1, 2, 4 (full); Modules 3 and 5 available in text/video format; AR/VR simulation components require in-vehicle or dealer format for completion	WCAG 2.1 AA compliance; screen reader compatible; closed captioning on all video; mobile-responsive; low-bandwidth mode available

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12. Operator and First Responder Training

12.1 Fleet Operator Certification

Commercial and governmental entities operating fleets of Space Cars carry enhanced compliance obligations that extend beyond those of individual PPs. Fleet operators have both the influence and the responsibility to establish organizational cultures of behavioral safety across their entire PP workforce.

Certification Requirements: All fleet managers responsible for Space Car operations must complete an 8-hour Advanced Doctrine Training program covering:

- Full Doctrine overview with emphasis on fleet-specific provisions (Section 1.2, Section 13, Section 15, Section 18)
- Fleet-wide BRI monitoring: understanding aggregate BRI data, identifying fleet-level behavioral risk trends, and using fleet reporting tools
- Incident response protocols: fleet manager responsibilities in Tier 2, 3, and 4 incident events (Section 18)
- Employee PP rights: understanding that employed PPs retain full individual data rights; employer restrictions on use of PP behavioral data for employment decisions
- Compliance reporting: quarterly self-reporting requirements, annual audit preparation, manufacturer interface for compliance documentation
- Driver wellbeing programs: best practices for addressing behavioral risk factors proactively through employee support programs

Reporting Obligations: Certified Fleet Operators are responsible for:

- Quarterly submission of aggregated fleet BRI metrics to the manufacturer compliance system
- Annual fleet-wide compliance report to the SVSC
- Immediate notification to the manufacturer within 12 hours of any Tier 3 or Tier 4 incident involving a fleet vehicle
- Maintaining documented training records for all fleet PPs

12.2 First Responder Integration Training

Law enforcement officers, emergency medical services personnel, and fire rescue teams may encounter Space Cars in any phase of the behavioral management system, including Hard Lock and Maximum Danger State. Untrained first responders interacting with a Hard Lock vehicle may inadvertently compromise the safety protocols or delay emergency response. The SVSC provides a certified First Responder Integration Training program, available at no cost, through the SVSC training portal.

Training Topics for Law Enforcement:

- Space Car vehicle identification: recognizing class markings, status light protocols (red strobing = BRI 75+), and external communication interfaces
- Safe Harbor approach protocol: approach vectors that do not trigger additional vehicle safety responses; proper positioning relative to the sealed vehicle
- Sealed vehicle protocols (BRI 100): how to initiate the biometric override procedure; who is authorized to perform biometric override; what data the vehicle will transmit upon law enforcement connection
- Escalation scenarios: how to interpret the vehicle's external status display; communication with the PP through the external speaker interface; use of the SVSC Law Enforcement Data Request portal for real-time BRI and ESC data

Training Topics for EMS:

- Medical emergency approach: how the vehicle signals a medical emergency event (external amber strobing); automated data transmission to EMS dispatch upon SEOF-02 (patient vitals, GPS location, ESC classification)
- Vehicle access in medical emergency: EMS biometric override for SEOF-02 events; sequence for safely opening vehicle without triggering additional security responses
- PP medical history access: how to query the vehicle's AI assistant for any PP-registered medical information (medications, allergies, emergency contacts); limitations of what the vehicle can disclose
- Pediatric response: Child-Alone protocol procedures and how EMS interacts with the vehicle upon Safe Harbor arrival

12.3 Manufacturer Service Training

Technicians who are authorized to access Space Car BRI and ESC logs in the context of maintenance, compliance review, or incident investigation must complete dedicated ethics and data integrity training.

- **Ethics Training (4 hours):** Mandatory for all technicians with PSA access. Covers the ethical obligations of handling sensitive behavioral and biometric data; the prohibition on accessing IPP data beyond the scope of the service request; the mandatory reporting requirement for any discovered evidence of abuse or tampering; and the personal liability of technicians for unauthorized data access.
- **Data Chain-of-Custody Procedures:** All BRI/ESC log access during service must be logged with technician identifier, timestamp, reason for access, and data accessed. The access log is maintained in tamper-evident storage and is available to the PP upon request and to regulatory authorities upon lawful demand. No technician may access BRI logs without a documented service request that specifically identifies the data access as necessary.
- **Certification:** Manufacturer service technicians must pass a Manufacturer Service Ethics Assessment (minimum 85% pass) before being granted PSA access credentials. Credentials expire annually and require re-certification.

Part VI

Manufacturer Compliance Specifications

13. Hardware Compliance Requirements

13.1 PSA Hardware Mandates

All Space Cars submitted for SVSC certification must incorporate a Psychometric Sensor Array meeting the following minimum hardware specifications. These specifications represent the floor of compliance — manufacturers are encouraged to exceed them wherever technically feasible.

Component	Specification Parameter	Minimum Requirement	Rationale
Multi-Spectral Camera (per zone)	Resolution	≥4K (3840×2160) per zone	Micro-expression detection requires sub-centimeter facial feature resolution at cabin distances
Multi-Spectral Camera (per zone)	Frame Rate	≥30 fps sustained; ≥120 fps burst capability	Micro-expressions (sub-200ms duration) require minimum 5 frames per expression
Biometric Seat Sensors	Baseline Biometric Match Accuracy	≥97% across validated demographic groups	Less than 97% accuracy creates unacceptable false-positive and false-negative rates in HRV and GSR monitoring
Biometric Seat Sensors	Data Latency	≤100ms from signal capture to BRI input	500ms BRI cycle requires all sensor inputs within 100ms of cycle start
Vocal Analysis Microphone Array	Analysis Latency	≤200ms from vocalization start to analysis output	SEOF-04 threat detection must operate within single BRI cycle
All Sensors	Data Encryption (in transit)	AES-256 minimum; TLS 1.3 for network transmission	PSA data is biometric data and must be protected at the highest practical

Component	Specification Parameter	Minimum Requirement	Rationale
			encryption standard
BRI Co-Processor	Processing Isolation	Dedicated hardware security module (HSM); physically isolated from infotainment and navigation processors	BRI computation must not be affected by other system load; security isolation prevents compromise of safety-critical calculations
All Safety-Critical Sensors	Operating Temperature Range	−40°C to +85°C; no degradation in accuracy within this range	Space Cars operate in extreme environmental conditions across all Climate zones

13.2 Redundancy Requirements

All safety-critical components of the Space Car must incorporate redundancy sufficient to maintain operational integrity during the failure of any single component. Redundancy requirements are specified at the system level:

- **Hot-Standby Backup Sensors:** All safety-critical PSA sensors must have a hot-standby backup in active parallel operation. "Hot-standby" means the backup sensor is actively collecting and processing data at all times, not merely dormant. In the event of primary sensor failure, the backup assumes operational status within 50ms with no interruption to BRI computation.
- **BRI Co-Processor Redundancy:** BRI computation must run on an isolated co-processor with a second-tier backup processor. If the primary co-processor fails, the backup must assume full BRI computation within 100ms. If both processors fail, Soft Lock must be engaged immediately — the system defaults to the most conservative safe state.

- **Failsafe Defaults:** Each sensor failure mode must have a defined failsafe behavior. Failsafe defaults are specified below:
    - Camera failure → ESC classification degrades to best-available from remaining sensors; BRI PSI contribution set to 35 (conservative mid-range) until camera restored; PP notified of degraded monitoring
    - Biometric seat sensor failure → KSI and PSI partially impaired; BRI thresholds reduced by 10 points (more conservative); PP notified
    - Microphone array failure → SEOF-04 disabled; vocal analysis unavailable; BRI PSI contribution set to 30; Naughty Course threshold reduced by 5 points; PP notified
    - Complete PSA failure → Immediate Soft Lock; PP explicitly notified; journey continued only in autonomous safe mode navigating to nearest service location or Safe Harbor

13.3 Safe Harbor Landing/Parking System

The Safe Harbor Navigation System is a mandatory, dedicated subsystem of the Space Car that enables autonomous navigation to designated Safe Harbor locations during Hard Lock and higher BRI events. The following hardware provisions are mandatory:

- **Emergency Propulsion Reserve:** A dedicated 20% battery/fuel reserve must be maintained exclusively for Safe Harbor navigation. This reserve is inaccessible to normal journey operation and can only be drawn upon by the Safe Harbor Navigation System upon Hard Lock engagement. The reserve must be sufficient to complete navigation to the nearest Safe Harbor from any point within the vehicle's rated operational range.
- **Auto-Door Lock on Hard Lock Engagement:** Upon Hard Lock engagement, all cabin doors automatically lock from the outside. Door locks may be released by: EMS or law enforcement biometric override; PP biometric identity confirmation

for non-SEOF Hard Lock events (following authorized de-escalation); or automatic release upon Safe Harbor arrival and EMS confirmation of scene safety.

- **External Status Light System:** All Space Cars must incorporate an external status light system visible from at minimum 50 meters in daylight conditions:
  - Standard operation: no status light active (unobtrusive)
  - Naughty Course Phase 3 (BRI 70–74): solid amber external light
  - Hard Lock (BRI 75+): red strobing at 2Hz
  - SEOF event: red strobing at 4Hz
  - Medical emergency (SEOF-02): amber strobing at 2Hz (distinguishes medical from behavioral emergency for first responder approach)
- **External Communication Interface:** A weatherproof external speaker and microphone system enabling communication between first responders and the PP inside a sealed vehicle, without requiring door opening. Interface must be accessible to first responders without special tools.

13.4 Vehicle Class-Specific Requirements

Requirement	Class I (Terrestrial)	Class II (Low-Altitude)	Class III (Transitional Orbital)
PSA Tier	PSA-1 (standard)	PSA-2 (enhanced + redundant)	PSA-3 (full redundancy + external environmental sensors)
External Sensor Suite	Not required	Required: basic airspace awareness (ADS-B, radar)	Required: full airspace awareness + weather + terrain + orbital interface sensors

Requirement	Class I (Terrestrial)	Class II (Low-Altitude)	Class III (Transitional Orbital)
Safe Harbor System	Ground Safe Harbor navigation	Ground + aerial Safe Harbor capability; must be able to execute emergency descent to Safe Harbor from any altitude within rated range	Ground + aerial Safe Harbor; plus orbital abort-to-Safe Harbor protocol for transition corridor operations
Emergency Propulsion Reserve	20% battery/fuel reserve	20% battery/fuel reserve; sufficient for Safe Harbor descent from maximum rated altitude	25% reserve (elevated for additional orbital corridor margin); dual-system backup propulsion required
Air Traffic Communication	Not required	Required: ATC communication capability; automatic position reporting in airborne phase	Required: ATC communication + orbital coordination authority communication; real-time position broadcast
BRI Co-Processor	Dual-processor (primary + backup)	Triple-processor (primary + backup + emergency-only tertiary)	Triple-processor with independent power supply for emergency-only tertiary

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14. Software and AI Compliance

14.1 BRI Algorithm Certification

The BRI computation algorithm is the analytical core of the Space Car's behavioral safety system, and its integrity and accuracy are non-negotiable. The SVSC

Technical Review Board (TRB) is the sole body authorized to certify BRI algorithms for use in production Space Cars.

Certification Requirements:

- **Algorithm Submission:** Manufacturers must submit full BRI algorithm documentation to the SVSC TRB at least 180 days before planned commercial launch. Documentation must include: mathematical specification of all sub-index calculations; all weighting factors and their derivation; all threshold values; all Behavioral Flag Increment values; all anti-gaming logic; and the complete sensor fusion methodology.
- **Explainability Requirement:** The BRI algorithm must be explainable — every BRI score must be decomposable into its contributing sub-indices, and every sub-index must be decomposable into its sensor inputs, in a manner that can be communicated to a non-technical PP in plain language. No black-box machine learning model may be the sole determinant of BRI. ML components may contribute to sub-index computation as one input among multiple, but the overall BRI calculation must follow the documented weighted formula with full explainability at every step.
- **Annual Re-Certification:** BRI algorithm certification expires annually. Manufacturers must submit an annual re-certification package including: any algorithm changes made in the prior year; updated bias audit results; real-world performance data; and SVSC TRB review of any anomalies flagged by the quarterly monitoring system.

14.2 Whisper Course Software Ethics Certification

The Whisper Course implementation requires an independent ethics certification process separate from the BRI algorithm certification. This process is conducted by the SVSC Ethics Board, not the TRB, reflecting the distinct ethical (rather than purely technical) nature of the assessment.

- **Calming-Without-Deception Compliance:** The Ethics Board will review the Whisper Course implementation against the ethical constraints defined in Section 6.4. Manufacturers must demonstrate through documented testing and independent audit that: (a) the Whisper Protocol interventions produce genuine physiological calming and not false biosignals that artificially depress the BRI; (b) all mandatory disclosures remain fully prominent during Whisper Course; (c) the system reliably exits Whisper Course under all defined exit conditions (Section 6.5); and (d) Whisper Course opt-out is fully functional and respected in all tested scenarios.
- **PP Experience Review:** The Ethics Board requires testimony or survey data from a minimum of 100 PP participants who consented to Whisper Course testing during development, with full disclosure of the testing purpose, regarding their experience of the interventions and whether any intervention felt coercive or distressing.
- **Annual Ethics Audit:** Post-market, the Whisper Course implementation is subject to an annual independent ethics audit, including analysis of real-world Whisper Course activation and outcome data and PP complaint records related to Whisper Course experiences.

14.3 OTA Update Protocol

Over-the-Air software updates to safety-critical systems require pre-approval from the SVSC TRB before deployment. The following protocols govern the OTA update process:

- **Safety-Critical Update Definition:** An OTA update is safety-critical if it modifies: BRI computation algorithms; ESC classification logic; threshold values; behavioral flag trigger conditions; SEOF detection logic; Whisper Course protocols; Hard Lock or Safe Harbor navigation systems; PSA data processing; or any other component directly affecting course assignment or emergency response.

- **Standard Approval Process:** Safety-critical OTA updates must be submitted to the SVSC TRB with full change documentation at least 60 days before planned deployment. The TRB issues approval, conditional approval (with required modifications), or rejection within 45 days. Approved updates may be deployed upon TRB notification of deployment intent.
- **Emergency Patch Process:** If a safety-critical vulnerability is discovered requiring immediate remediation, the manufacturer may request emergency patch approval. The TRB convenes an emergency panel within 24 hours of notification. Emergency patch approval is granted within a maximum 72-hour window. Manufacturers may deploy a temporary non-OTA mitigation (e.g., reduced operational envelope) during the 72-hour review period.
- **Mandatory Rollback Capability:** All OTA updates to safety-critical systems must include a validated rollback procedure that can restore the prior software state within 4 hours of rollback initiation. Rollback capability must be maintained for the two most recent prior software versions. Manufacturers must successfully test rollback procedures before any update is deployed to the fleet.

14.4 Data Architecture Requirements

The data architecture governing BRI logs, ESC records, and IPP data must comply with the following mandatory requirements:

- **Encryption at Rest:** All BRI log data, ESC classification records, and PSA sensor data stored locally on the vehicle must be encrypted using AES-256 or equivalent. Encryption keys must be stored in a hardware security module (HSM) separate from the data storage medium.
- **Encryption in Transit:** All data transmitted from the vehicle to manufacturer cloud systems, regulatory authorities, or third parties must be encrypted using TLS 1.3 or equivalent. Unencrypted transmission of BRI or biometric data in any form is prohibited.

- **Minimum Local Retention — 90 Days:** All BRI event logs and associated sensor snapshots must be retained in local encrypted storage for a minimum of 90 days from the date of the event. Local data may not be deleted before the 90-day minimum absent PP explicit deletion request and confirmation that no active dispute or regulatory investigation covers the data.
- **Minimum Cloud Retention — 2 Years:** BRI event logs at Phase 2 severity or higher must be transmitted to the manufacturer's secure cloud storage and retained for a minimum of 2 years from the date of the event for regulatory access purposes. Cloud data is subject to the same encryption requirements as local data.
- **PP Data Sovereignty Tools:** The vehicle's UI must include, as a native, always-accessible feature:
    - IPP data access portal: PP can view their full IPP data at any time
    - BRI history viewer: visual display of BRI trajectory for any journey in the retention window
    - Consent management center: PP can review and modify all data sharing consents
    - Data export request: one-click initiation of full IPP data export request
    - Dispute initiation: direct access to Threshold Dispute filing

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15. Compliance Testing and Certification

15.1 Pre-Market Testing Protocol

Before any Space Car model is authorized for commercial sale, the manufacturer must complete and document the following pre-market testing protocol. Testing results are submitted to the SVSC TRB as part of the certification application package.

Test Category	Description	Required Volume	Pass Criterion
BRI System Simulation	Full BRI computation simulation across synthetic PP profiles spanning the full demographic and behavioral range of the intended PP population	Minimum 10,000 synthetic PP profiles; minimum 100 journey scenarios per profile segment	BRI computation accuracy within ±2 points of ground truth across 99.5% of test cases; no demographic bias exceeding 5% between groups
Whisper Course Ethics Stress-Test	Activation, intervention, and exit logic testing under adversarial scenarios designed to probe for ethical constraint violations	Minimum 500 adversarial test scenarios covering all six exit conditions and all four ethical constraints	Zero failures of mandatory ethical constraints; all exit conditions trigger correctly in 100% of applicable scenarios
Hard Lock Reliability Test	Hard Lock activation testing across all trigger conditions (BRI threshold, SEOF) under varied environmental and technical conditions	Minimum 1,000 test activations across all vehicle systems states (all sensor combinations, all weather conditions, primary/backup processor states)	Hard Lock activation success rate ≥99.99% (maximum 1 failure per 10,000 test activations)
Safe Harbor Navigation Accuracy	Navigation accuracy testing for autonomous Safe Harbor transit under Hard Lock conditions	Minimum 500 navigation tests across varied geographies, weather conditions, and traffic scenarios	Arrival within ≤50 meters of target Safe Harbor landing/parking position in ≥99.5% of tests
Sensor Failure Failsafe Testing	Validation of all defined failsafe	All defined failure modes tested	Correct failsafe default engaged

Test Category	Description	Required Volume	Pass Criterion
	default behaviors under simulated sensor failure conditions	across all vehicle classes; each failure mode tested minimum 100 times	within defined latency requirements in 100% of test cases
ESC Demographic Accuracy	ESC classification accuracy testing across the full demographic range of the intended PP population	Testing with minimum 200 human participants across validated demographic groups; minimum 50 per group	≥94% overall accuracy; ≥92% per demographic group; no systematic bias pattern across groups

15.2 Ongoing Compliance Monitoring

SVSC certification is not a one-time event — it is an ongoing obligation maintained through continuous reporting and periodic audit. The following monitoring requirements apply to all certified manufacturers:

- **Quarterly Self-Reported Metrics:** Manufacturers must submit quarterly compliance reports to the SVSC covering: total fleet journey count; BRI event distribution by threshold tier; Naughty Course activation rate; Whisper Course activation rate and outcome distribution; Hard Lock rate; SEOF event count by SEOF code; Safe Harbor navigation activations; and PP dispute count and resolution rate.
- **Annual Third-Party Audit:** Each certified manufacturer must commission an independent third-party compliance audit annually. The audit must be conducted by an SVSC-accredited audit firm (not affiliated with the manufacturer) and must assess: BRI algorithm performance against certified specification; ESC classification accuracy; bias audit results; data architecture compliance; OTA update compliance; PP training delivery compliance; and incident reporting completeness. Audit reports are submitted to the SVSC TRB within 30 days of completion.

- **Incident Reporting:** Within 24 hours of any BRI 90+ event (Full System Override or Maximum Danger State), the manufacturer must transmit a preliminary incident notification to the SVSC. A full incident report is required within 72 hours. See Section 18 for complete incident reporting requirements.

15.3 Compliance Failure Consequences

Violation Category	Examples	First Offense	Second Offense (within 24 months)	Third Offense or Critical Failure
Reporting Failure	Late quarterly report; incomplete incident notification; missing annual audit	Written warning; mandatory corrective action plan within 30 days	Fine: $50,000 per instance; corrective action verification audit	Fine: $250,000; public notice of non-compliance; enhanced monitoring for 12 months
Algorithm Non-Compliance	BRI algorithm modified without TRB approval; explainability requirement violated; bias audit failure unremediated	Mandatory algorithm review and remediation within 90 days; enhanced monitoring	Fine: $500,000; fleet operational restriction pending remediation	Certification suspension; fleet grounding for non-compliant models; mandatory recall of affected units pending remediation
Ethical Constraint Violation	Whisper Course used to suppress mandatory warning; emergency request overridden; ESC-7 mishandled	Immediate certification suspension; emergency audit within 30 days; remediation required before reinstatement	Certification revocation; fleet grounding; SVSC public safety notice issued	Permanent certification revocation; referral to law enforcement; civil liability
Hardware Non-Compliance	PSA sensor below minimum	Mandatory recall of affected units;	Fine: $1,000,000; expanded	Certification revocation; full fleet

Violation Category	Examples	First Offense	Second Offense (within 24 months)	Third Offense or Critical Failure
	specification; redundancy system failure undisclosed; Safe Harbor reserve compromised	corrective hardware update required	recall; enhanced inspection protocol	recall; referral to NHTSA/FAA for defect investigation
Data Breach or Privacy Violation	Unencrypted PP data transmission; unauthorized third-party data sharing; IPP data access without authorization	Mandatory breach notification; investigation; corrective action plan	Fine: $2,000,000; mandatory independent privacy audit; enhanced monitoring	Certification revocation; referral to FTC; potential criminal referral for deliberate violations

Part VII

Policy and Legal Framework

16. Regulatory Integration

16.1 Alignment with Existing Frameworks

The Space Car Safety Doctrine does not replace existing regulatory authority — it operates as a purpose-built behavioral safety supplement to the constellation of federal frameworks that collectively govern Space Car operations. The following alignment provisions define how this Doctrine interfaces with each primary regulatory framework:

- **FAA UAM/UAS Regulations:** The FAA's Urban Air Mobility Integration Pilot Program and associated 14 CFR amendments govern all airspace aspects of Space Car Class II and III operations. This Doctrine's behavioral safety provisions apply to the PP during all airborne phases, with specific provisions (air corridor suspension at Naughty Course Phase 2; emergency descent protocol for Hard Lock; Class II/III Safe Harbor aerial navigation) designed to complement FAA airspace management requirements without conflicting with ATC authority.
- **NHTSA Autonomous Vehicle Guidelines:** The NHTSA's Automated Driving System (ADS) framework and Federal Motor Vehicle Safety Standards (FMVSS) apply to all terrestrial operations. This Doctrine's SAE J3016 autonomy level references are aligned with NHTSA's level definitions. BRI-driven course management operates within the autonomy architecture permitted by NHTSA ADS guidelines.
- **FCC RF Spectrum Rules:** PSA sensor systems that employ RF-based signals (including radar for external environment sensing, Bluetooth/ANT+ for wearable integration, and cellular connectivity for cloud data transmission) must comply with FCC Part 15, Part 18, and applicable licensed spectrum rules. RF emissions from PSA systems must not interfere with aviation communications, ATC radar, or emergency services RF systems.
- **ISO 26262:2018 Functional Safety:** All safety-critical electronic systems — including BRI co-processors, Hard Lock trigger systems, Safe Harbor Navigation System, and PSA sensor processing modules — must achieve a minimum of ASIL-B (Automotive Safety Integrity Level B) overall, with ASIL-D required for the BRI computation engine and Hard Lock trigger systems. ISO 26262 certification is a prerequisite for SVSC certification and must be demonstrated through a certified ISO 26262 functional safety management process.

- **SAE J3016 Autonomy Levels:** Space Car autonomy levels are defined by SAE J3016 as follows: Class I vehicles operating terrestrially must meet SAE Level 3 minimum (conditional automation); Class II vehicles in airborne phase must meet SAE Level 4 minimum (high automation) during airborne transit; Class III vehicles in transitional corridor must meet SAE Level 4 for the full transition sequence. BRI-driven Hard Lock and Full System Override events represent Level 4+ (effectively Level 5 — full automation) behavioral safety responses.

16.2 Jurisdictional Applicability

This Doctrine is established as the primary behavioral safety standard for the United States domestic market, with the following jurisdictional framework:

- **Federal Domestic Application:** This Doctrine applies in all 50 states, the District of Columbia, and US territories for all Space Car operations within US airspace and on US surface infrastructure. States may enact supplementary behavioral safety provisions that are more stringent than this Doctrine but may not enact provisions that conflict with or weaken the minimum standards established herein.
- **International Harmonization — EASA:** The European Union Aviation Safety Agency's (EASA) Advanced Air Mobility (AAM) regulatory framework shares significant structural overlap with the SVSC Doctrine. The SVSC maintains an active harmonization working group with EASA to develop mutual recognition arrangements for BRI algorithm certification and PSA hardware standards. PPs with SVSC-certified Space Cars operating in EU jurisdictions that have adopted EASA AAM framework must also comply with any additional EU-specific requirements.
- **International Harmonization — Transport Canada:** A bilateral Mutual Recognition Agreement (MRA) covering PSA hardware specifications and BRI certification is under active negotiation with Transport Canada (TC). Upon

execution, SVSC-certified hardware meeting PSA specifications will be accepted by TC without duplicate certification.

- **International Harmonization — ICAO:** The International Civil Aviation Organization's Standards and Recommended Practices (SARPs) for Urban Air Mobility are being developed under the ICAO DRONE ENABLE program. The SVSC participates in ICAO DRONE ENABLE working groups and will update this Doctrine to incorporate adopted SARPs as they are published.

16.3 SVSC Authority and Governance

The Space Vehicle Safety Consortium is governed by a multi-stakeholder board structure designed to ensure that no single interest group — manufacturer, government, or public — can unilaterally direct Doctrine development or certification outcomes.

Governing Body	Composition	Primary Authority	Decision Method
Technical Review Board (TRB)	9 members: 3 engineering/AI experts, 2 behavioral scientists, 2 regulatory liaison members (FAA, NHTSA), 2 independent technologists. Terms: 3 years staggered.	BRI algorithm certification; OTA update approval; hardware specification maintenance; technical dispute resolution	Super-majority (6/9) for certification; simple majority for technical guidance
Ethics Board	7 members: 2 bioethicists, 1 clinical psychologist, 1 civil rights attorney, 1 privacy law specialist, 1 disability rights advocate, 1 consumer	Whisper Course ethics certification; bias audit review; PP rights policy; cultural variation standards	Consensus preferred; super-majority (5/7) for certification decisions

Governing Body	Composition	Primary Authority	Decision Method
	protection representative. Terms: 3 years staggered.		
PP Advocacy Panel	5 members: 3 elected by SVSC-registered PP population, 1 appointed by consumer advocacy coalition, 1 appointed by disability rights coalition. Terms: 2 years.	PP dispute escalation review; IPP privacy policy input; training program adequacy review; public communication of PP rights	Simple majority; all decisions advisory except PP dispute resolutions which are binding on manufacturers
Manufacturer Liaison Committee	One non-voting representative from each SVSC-certified manufacturer. Rotating chair.	Industry input on proposed amendments; compliance challenge notification; voluntary early disclosure of compliance concerns	Advisory only; no voting authority in Doctrine amendment
Law Enforcement Liaison	One representative each from: National Sheriffs' Association, International Association of Chiefs of Police, International Association of Fire Chiefs, EMS practitioners' association. Rotating chair.	First responder protocol input; SEOF design review; Safe Harbor location standards input; Hard Lock resolution protocol design	Advisory; binding authority on first responder protocol provisions subject to board confirmation
SVSC Governing Board	Chairs of each above body plus SVSC Executive Director. Quorum: 4 of 6.	Doctrine amendment approval; certification appeals; emergency	Super-majority (4/6) for Doctrine amendments; simple majority for operational decisions

Governing Body	Composition	Primary Authority	Decision Method
		amendments; annual budget and strategic direction	

17. Liability and Legal Protections

17.1 Liability Allocation Matrix

The following matrix allocates primary liability responsibility for Space Car incidents across the key categories of causation. Liability allocation does not preclude the assignment of secondary or contributory liability; it establishes the primary party bearing the burden of proof and remediation in the first instance.

Incident Category	Causation Description	Primary Liable Party	Secondary Party	Doctrine Reference
PP-Triggered Event	Incident caused by PP behavioral actions (e.g., aggressive driving, substance impairment, deliberate system circumvention) where BRI and ESC data confirms PP behavioral origin	Pilot-Passenger	Manufacturer (if PSA failure contributed); Fleet Operator (if training failure contributed)	Section 5, Section 7, Section 8
System-Triggered	Incident where	No strict liability —	Manufacturer if system	Section 17.2

Incident Category	Causation Description	Primary Liable Party	Secondary Party	Doctrine Reference
Event	automated system action (course assignment, Hard Lock, Safe Harbor navigation) is the proximate cause of harm and was triggered correctly per doctrine protocols	Good Faith Compliance Shield applies (Section 17.2)	acted within specification; PP bears proportional responsibility for triggering BRI state	
Sensor Failure Event	Incident caused by PSA sensor failure producing incorrect BRI/ESC output that drove an incorrect system response	Manufacturer (design or manufacturing defect); Supplier (component defect)	Fleet Operator if maintenance failure contributed; Third-party service provider if improper service caused failure	Section 13.1, Section 13.2
Manufacturer Defect Event	Incident caused by a hardware or software defect not attributable to sensor failure — e.g., BRI algorithm error, OTA update defect, Safe Harbor navigation failure	Manufacturer	SVSC (if certification process failed to identify defect); Software supplier if defect originated in third-party component	Section 14, Section 15
Third-Party Interference Event	Incident caused by deliberate third-party interference with the	Third-Party Interfering Actor (criminal liability)	Manufacturer (if cybersecurity deficiency contributed); Manufacturer	Section 13.1 (encryption requirements)

Incident Category	Causation Description	Primary Liable Party	Secondary Party	Doctrine Reference
	Space Car's systems — hacking, signal jamming, physical tampering		bears obligation to assist investigation	
Training Failure Event	Incident attributable to demonstrable gap in PP training — PP was not adequately informed of a critical protocol — where training records show non-completion or delivery failure	Fleet Operator (if PP was employed fleet driver); Manufacturer (if training delivery system failed)	PP if training was available but not completed; SVSC if training content was inadequate	Section 11, Section 12

17.2 Good Faith Compliance Shield

Manufacturers and operators who can demonstrate documented, good-faith compliance with this Doctrine at the time of an incident are protected by the Good Faith Compliance Shield — a liability protection provision that prevents strict liability claims arising solely from the operation of the Space Car's behavioral safety systems when those systems functioned as designed and certified.

The Good Faith Compliance Shield applies when:

19. The manufacturer holds valid, current SVSC certification for the vehicle model involved;

20. The specific system functions involved in the incident (BRI computation, course assignment, Hard Lock, Safe Harbor navigation) have been certified by the SVSC TRB;
21. The vehicle's actual system behavior at the time of the incident is consistent with the certified specification;
22. The manufacturer has no outstanding non-remediated compliance failures identified by the SVSC at the time of the incident; and
23. No evidence of deliberate misrepresentation to the SVSC during certification exists.

The Good Faith Compliance Shield does not protect against:

- Negligence liability for known defects unreported to the SVSC;
- Strict product liability where a manufacturing defect (distinct from design) caused the harm;
- Claims arising from violations of the ethical constraints in Section 6.4;
- Data privacy violations under applicable state or federal privacy law.

17.3 PP Rights and Protections

The following rights are held by all Pilot-Passengers and are not waiveable through Terms of Service agreements, registration agreements, or any other contractual mechanism:

- **Right to Audit BRI History:** The PP has the right to access their complete BRI history for all journeys within the data retention window, in a human-readable format, through the vehicle UI or manufacturer PP Portal, at any time and without cost.
- **Right to Challenge Automated Decisions:** The PP has the right to formally contest any automated threshold decision (course assignment, Hard Lock, SEOF trigger) through the Threshold Dispute process (Section 8.4) within 30 days of the triggering event.

- **Protection from Discriminatory Profiling:** The PP is protected from behavioral profiling based on demographic characteristics (race, gender, age, disability, religion, national origin, sexual orientation) as specified in Section 10.4. Any PP who has reason to believe they are experiencing discriminatory profiling may file a Bias Complaint with the SVSC Ethics Board without filing through the manufacturer's PP Portal.
- **Right to Human Review:** Automated decisions resulting in Hard Lock or higher that the PP contests have the right to human review — not merely algorithmic review — by the manufacturer's compliance team within 30 business days and by the SVSC PP Advocacy Panel upon escalation.
- **Right to Informed Withdrawal:** A PP who no longer wishes to operate a Space Car under the SVSC behavioral monitoring framework has the right to terminate use of the vehicle and request full deletion of their IPP, subject only to retention obligations imposed by active regulatory investigations or court orders.
- **Right to Counsel:** In any proceeding before the SVSC — whether a Threshold Dispute, Bias Complaint, or incident investigation — the PP has the right to be represented by legal counsel of their choosing.

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18. Incident Classification and Reporting

18.1 Incident Tier Table

Tier	BRI Range at Peak	Injury/ Property Damage	Law Enforcement Involved?	Resolution	Classification Name
Tier 1	BRI 50–74	No injury; no property damage beyond incidental	No	Self-resolved through de-escalation protocol; vehicle returned to Peaceful Course	Elevated-Caution Event
Tier 2	BRI 50–74 with Phase 3 engagement, OR BRI 75–89 de-escalated before injury	No serious injury; minor property damage possible	Possible; law enforcement may have been notified but no active law enforcement intervention	Resolved through Safe Harbor navigation or de-escalation with EMS pre-staging only	Significant Safety Event
Tier 3	BRI 75–89 with EMS dispatch, OR BRI 90–99	Minor to moderate injury; significant property damage	Yes — law enforcement response occurred	Incident required active emergency services intervention; vehicle at Safe Harbor with law enforcement	Serious Safety Incident
Tier 4	BRI 90–100 (Full System Override or Maximum Danger State)	Serious injury, critical injury, fatality, or catastrophic property damage	Yes — active law enforcement investigation opened	Incident requires full investigation; vehicle sealed; law enforcement in active investigation; SVSC independe	Critical Safety Event

Tier	BRI Range at Peak	Injury/ Property Damage	Law Enforcement Involved?	Resolution	Classification Name
				nt review mandatory	

18.2 Mandatory Reporting Requirements by Tier

- **Tier 1 Events:** Logged internally in the vehicle's encrypted event log and transmitted to the manufacturer's compliance system in the next scheduled data sync. No external reporting required. Aggregate Tier 1 data reported in quarterly compliance metrics.
- **Tier 2 Events:** Manufacturer must generate a Tier 2 Event Report within 72 hours of the event. The report must include: event timestamp and duration; BRI trajectory throughout the event; ESC classifications at each transition; trigger conditions; interventions applied; outcome; and PP de-escalation engagement data. Reports are retained in the manufacturer's compliance system and are available to the SVSC upon request.
- **Tier 3 Events:** Manufacturer must transmit a preliminary Tier 3 Incident Notification to the SVSC within 24 hours of the event. A full Tier 3 Incident Report is required within 7 days. The SVSC TRB reviews all Tier 3 incidents for systemic patterns. If three or more Tier 3 incidents share a common cause factor, the SVSC may initiate a systemic safety investigation against the manufacturer.
- **Tier 4 Events:** Manufacturer must transmit immediate notification to both the SVSC and the relevant law enforcement jurisdiction upon event detection. Notification must include: vehicle identification, GPS location, current BRI and ESC state, SEOF code if applicable, and a real-time data link for law enforcement access to BRI/ESC logs. A full Tier 4 Critical Event Report is

required within 24 hours. The SVSC convenes an Independent Review Panel for all Tier 4 events.

18.3 Incident Investigation Protocol

Tier 3 and Tier 4 events are subject to formal investigation under the SVSC Incident Investigation Protocol:

- **Chain of Custody for BRI/ESC Log Data:** Upon a Tier 3 or Tier 4 event, the vehicle's BRI/ESC log data and all associated sensor snapshots must be preserved under a documented chain of custody protocol. The manufacturer must generate a cryptographic hash of the log data at the time of event detection and transmit the hash to the SVSC within 2 hours. This hash serves as the integrity reference for all subsequent evidence use. Any modification of log data after hash generation constitutes evidence tampering and triggers automatic Certification Suspension.
- **Independent Review Panel for Tier 3 and 4 Events:** The SVSC convenes an Independent Review Panel (IRP) for all Tier 3 events involving serious injury and all Tier 4 events. The IRP consists of three members: one SVSC TRB member, one SVSC Ethics Board member, and one independent subject matter expert selected from the SVSC roster. The IRP operates independently of the manufacturer and has full access to all BRI/ESC log data. The IRP issues a Finding Report within 60 days (Tier 3) or 90 days (Tier 4) of the event, with findings on causation, liability allocation, and required corrective actions.
- **PP Participation in Investigation:** The PP involved in a Tier 3 or Tier 4 incident has the right to participate in the IRP process — to provide testimony, to be represented by counsel, and to receive a copy of the IRP's Finding Report. The PP's participation is voluntary.

19. Policy Version Control and Amendment

19.1 Current Version

This document represents **Policy Version 1.0** of the Space Car Safety Doctrine — Unified Edition, effective January 1, 2026. Version 1.0 is the inaugural edition of the Doctrine and establishes the foundational framework upon which all subsequent versions will build. The SVSC commits to maintaining this Doctrine as a living document, updated to reflect technological advancement, operational experience, and evolving best practices in behavioral safety and AI ethics.

19.2 Standard Amendment Process

Proposed amendments to this Doctrine may be submitted by any of the following parties: SVSC Governing Board members, SVSC TRB or Ethics Board, Certified Manufacturers (through the Manufacturer Liaison Committee), PP Advocacy Panel, recognized regulatory agencies (FAA, NHTSA, FCC), or members of the public (through the SVSC Public Comment portal). All amendments follow the standard process:

24. **Submission:** Proposed amendment submitted to the SVSC Technical Committee in writing, with full justification, supporting evidence, and proposed implementation timeline.
25. **60-Day Public Comment Period:** The proposed amendment is published on the SVSC public website. Written comments accepted from any party. SVSC staff summarizes and categorizes all public comments for Governing Board review.
26. **Ethics Board Review:** The Ethics Board reviews the proposed amendment for compliance with the ethical principles underlying this Doctrine, with

specific attention to any amendment affecting the Whisper Course, PP data rights, or bias protection provisions.

27. **Technical Review Board Review:** The TRB reviews the proposed amendment for technical feasibility, implementation timeline realism, and compliance with applicable regulatory frameworks.
28. **Governing Board Vote:** The Governing Board convenes to review the full amendment package (proposal + public comments + Ethics Board opinion + TRB opinion) and votes. Super-majority (4/6) required for adoption.
29. **180-Day Implementation Window:** Adopted amendments take effect 180 days after Governing Board vote. Manufacturers are required to achieve compliance with the amended provision within this window or apply for a compliance extension (maximum additional 90 days, subject to Governing Board approval).

19.3 Emergency Amendment Process

Where a safety-critical deficiency in this Doctrine is identified that poses an immediate and material risk to PP or public safety, the SVSC may initiate an Emergency Amendment through the following expedited process:

30. The SVSC Executive Director, in consultation with the TRB Chair, determines that an emergency amendment is warranted and publishes an Emergency Safety Notice on the SVSC website.
31. An Emergency Amendment Working Group (EAWG) — comprising TRB Chair, Ethics Board Chair, one PP Advocacy Panel member, and one law enforcement liaison — convenes within 48 hours.
32. A 14-day expedited comment period is open to manufacturers and regulatory agencies (general public comment is not required in the 14-day window, but is solicited in a subsequent 30-day retroactive comment period).
33. The EAWG issues a final Emergency Amendment recommendation within the 14-day window.

34. The Governing Board votes by electronic ballot within 72 hours of EAWG recommendation. Simple majority sufficient for Emergency Amendment adoption.

35. Emergency Amendments take effect immediately upon adoption. Manufacturers receive a 30-day implementation window unless the emergency requires immediate action, in which case an Operational Guidance Note is issued covering interim requirements pending full implementation.

19.4 Version History Table

Version	Effective Date	Primary Changes	Adoption Vote
1.0	01 January 2026	Inaugural edition — full doctrine established	SVSC Governing Board — Unanimous (6/6) — 12 August 2025
1.1	TBD	[Reserved for future amendments]	—
2.0	TBD	[Reserved for major revision following Version 1 operational period]	—

Part VIII

Appendices

Appendix A — BRI Scoring Reference Card

This reference card provides the complete BRI scoring table for field reference by first responders, manufacturers, and fleet operators. For full system context, refer to Section 7.3 and Section 8.2.

BRI Score	ESC State	Course/ State	Speed Restriction	Destination Restriction	Emergency Status	PP Notified?
0–15	ESC-1 Serene	Peaceful (Reward)	None	None	None	No
16–25	ESC-2 Focused	Peaceful	None	None	None	No
26–40	ESC-3 Mildly Elevated	Peaceful (Enhanced Monitor)	None	None	None	No
41–49	ESC-4 Stressed	Peaceful (Critical Monitor)	None	None	None	No
50–59	ESC-4 Stressed	Naughty Course Phase 1	Limit +10%	None	None	No (Phase 1)
60–69	ESC-5 Agitated	Naughty Phase 2 / Whisper	Exact posted limit	Safe categories only	Emergency contact pre-alert	Yes (Phase 2)
70–74	ESC-6 Volatile	Naughty Phase 3 — Soft Lock	25 mph maximum	Safe Harbor only	EMS pre-staged	Yes — explicit
75–89	ESC-6/7	Hard Lock	Autonomous control	Safe Harbor mandatory	EMS dispatched; law enforcement notified	Yes — emergency notification

BRI Score	ESC State	Course/ State	Speed Restriction	Destination Restriction	Emergency Status	PP Notified?
90–99	ESC-7 Crisis	Full System Override	All PP inputs disabled	Safe Harbor mandatory	All services dispatched; law enforcement active	Yes — maximum alert
100	ESC-7 Crisis (Max)	Maximum Danger State	Stopped at Safe Harbor	Sealed at Safe Harbor	Law enforcement seal; full data transmission	Yes — sealed vehicle notification

Appendix B — Three-Course Decision Flowchart (Narrative Description)

Step 1 — Journey Initiation: PP enters vehicle. Pre-departure biometric scan (30 seconds). IPP retrieved. Initial BRI computed from available sub-indices. If no IPP exists, population-average baseline applied; calibration period begins.

Step 2 — Initial Course Assignment: IF BRI < 25 AND ESC ∈ {ESC-1, ESC-2} → ASSIGN Peaceful Course. IF BRI 25–49 → ASSIGN Peaceful Course with enhanced monitoring. IF BRI ≥ 50 OR Naughty-Course Flag detected → PROCEED to Step 3.

Step 3 — Naughty Course Phase Determination: IF BRI 50–59 → Phase 1. IF BRI 60–69 → Phase 2; evaluate for Whisper Course (Step 4). IF BRI 70–74 → Phase 3 (Soft Lock). IF BRI ≥ 75 → Hard Lock (Step 5). IF SEOF detected → Hard Lock immediately (Step 5).

Step 4 — Whisper Course Evaluation: IF Phase 2 active AND (ESC flag = AoN OR RDI) AND no Whisper Course contraindication → ACTIVATE Whisper Course in parallel. ELSE → Phase 2 explicit protocols only.

Step 5 — Hard Lock / Full System Override: IF BRI ≥ 75 → Hard Lock; autonomous navigation to Safe Harbor; EMS dispatched. IF BRI ≥ 90 → Full System Override; all PP inputs disabled; law enforcement notified. IF BRI = 100 → Maximum Danger State; vehicle sealed at Safe Harbor.

Step 6 — Continuous Loop: At each 500ms cycle (200ms if BRI ≥ 50), return to Step 2 data collection. Monitor for de-escalation (BRI decay) or escalation (BRI increase or new flag). Adjust Course Path assignment accordingly. Log all transitions. Continue until journey complete or Hard Lock resolved.

Appendix C — Whisper Course Intervention Menu

Intervention Code	Intervention Name	Mechanism	Implementation Rate	Ethical Constraint
WC-01	Binaural Relaxation Frequency Overlay	Low-frequency audio overlay (40–60 Hz) mixed into existing audio at −30 dB, increasing to −15 dB over 5 minutes	Gradual over 5 minutes	Must not interfere with emergency audio alerts; volume must not exceed safe hearing levels; must be fully disclosed in PP training
WC-02	Ambient Sound Layer Introduction	Low-level ambient sound (rain, ocean, instrumental) introduced at −20 dB when no audio playing	Immediate; gradual fade-in over 2 minutes	Must not mask emergency alerts; PP may override by selecting own audio content
WC-03	Micro-Acceleration Smoothing	Reduction of propulsion acceleration variance by	15% initial; 40% maximum over 3	Must not alter journey time by more than 5%; must not

Intervention Code	Intervention Name	Mechanism	Implementation Rate	Ethical Constraint
		15–40% over 3 minutes	minutes	prevent emergency acceleration/deceleration if required for collision avoidance
WC-04	Lighting Gradient (Luminance)	Gradual reduction in cabin luminance at 3–5% per minute toward 70% of current level	3–5% per minute; maximum reduction 30% from current setting	Must maintain minimum safety luminance (50 lux minimum); PP may override by requesting lighting adjustment
WC-05	Lighting Gradient (Color Spectrum)	Gradual shift in color temperature toward 5,000–5,500K (blue-green calm spectrum) at maximum 50K per minute	Maximum 50K per minute; full transition over 10–15 minutes	Same as WC-04; must not produce strobing or abrupt color changes
WC-06	Route Softening	Gradual heading adjustment toward Safe Harbor via natural route complexity without explicit destination change announcement	Applied at natural navigation decision points; maximum 2° per navigation decision point	MUST disclose Safe Harbor navigation if PP explicitly queries destination; MUST NOT mask emergency navigation change; applies only within window of normal route decision-making
WC-07	Conversational Deflection	AI assistant topic bridging	Applied in real time during	AI must not refuse to

Intervention Code	Intervention Name	Mechanism	Implementation Rate	Ethical Constraint
		from agitating to neutral/grounding subjects using open-ended questions and validation statements	PP-initiated conversation	discuss any topic; AI must not deny distressing facts the PP has raised; deflection is gradual redirection, not suppression; AI must respond truthfully to direct factual questions about the vehicle's status
WC-08	Temperature Comfort Adjustment	Gradual temperature adjustment toward PP's IPP-documented calming temperature range (±2°C maximum deviation from current set temperature)	Maximum 0.5°C per minute; maximum ±2°C total adjustment	PP may override via climate control at any time; adjustment must not cause physical discomfort

Appendix D — PSA Sensor Technical Specifications

For complete PSA hardware specifications, refer to Section 13.1 (mandatory specifications) and Section 10.2 (functional specifications). The following additional technical parameters apply to all PSA components:

- **Electromagnetic Compatibility (EMC):** All PSA components must comply with FCC Part 15 Class B for unintentional radiators and must not produce RF interference affecting aviation communications, ATC radar, or emergency services communications. EMC testing must be conducted per ANSI C63.4 for terrestrial environments and per RTCA DO-160 for avionics-adjacent environments (Class II and III vehicles).
- **Cybersecurity:** All PSA components with network connectivity must comply with NIST Cybersecurity Framework v2.0. PSA data must be transmitted only over authenticated, encrypted channels. PSA components must not accept firmware updates except through the SVSC-approved OTA update protocol (Section 14.3). Penetration testing by an SVSC-accredited cybersecurity firm must be conducted annually.
- **Calibration:** PSA sensors must be recalibrated at intervals specified by the manufacturer but not less than annually or every 50,000 miles (whichever comes first). Calibration must be performed by certified technicians using SVSC-approved calibration procedures. Calibration records are maintained in tamper-evident local storage.
- **Environmental Durability:** All PSA components must meet IP67 (ingress protection) minimum for cabin-installed components and IP69K for externally mounted components (Class II/III vehicles). Vibration resistance per SAE J1455. Operating temperature −40°C to +85°C without performance degradation.

Appendix E — Safe Harbor Zone Standards and Designation Criteria

A location may be designated as a Safe Harbor Zone by the SVSC upon application by a property owner, local government, or public authority. The following minimum criteria must be met for Safe Harbor Zone designation:

- **Physical Size:** Minimum landing/parking area of 10m × 10m clear of overhead obstructions (for Class II/III aerial Safe Harbor); minimum 6m × 12m for Class I ground Safe Harbor.
- **Emergency Services Access:** Must be within 5 minutes of a law enforcement station and within 10 minutes of an EMS station during all hours of operation.
- **Perimeter Security:** Physical perimeter adequate to prevent unauthorized pedestrian access to the vehicle during a sealed-vehicle event.
- **Communication Infrastructure:** Reliable cellular and/or Wi-Fi connectivity sufficient for SVSC data transmission during a Hard Lock event.
- **Lighting (Ground Safe Harbors):** Minimum illumination level of 100 lux during all hours of operation (active lighting required for nighttime designation).
- **24/7 Availability:** Must be accessible to Space Car navigation 24 hours per day, 7 days per week. Temporary closures (construction, events) must be reported to the SVSC within 24 hours for Safe Harbor database update.
- **Designation Process:** Application submitted to SVSC; site inspection conducted by SVSC-accredited inspector; designation granted within 60 days of complete application; annual renewal required; designation may be revoked if criteria are no longer met.

Appendix F — PP Training Curriculum Outline

For complete training program details, refer to Section 11. The following is a condensed curriculum outline for manufacturer training system implementation:

Module	Duration	Key Topics	Assessment Pass Threshold	Delivery Format Options
Module 1 — System Orientation	2 hours	Space Car operations, PSA sensors, BRI explanation, PP rights, IPP overview, 90-day calibration	80%	In-vehicle AR, Dealer VR, Online Portal
Module 2 — Course Awareness	1 hour	Three-Course Framework, Peaceful Course, Naughty Course phases, Whisper Course (full disclosure), voluntary de-escalation tools	80%	In-vehicle AR, Dealer VR, Online Portal
Module 3 — Emergency Protocols	1.5 hours	Hard Lock procedures, Safe Harbor navigation, emergency contact setup, medical emergency response, vehicle sealing, bystander guidance	80% (simulation component)	In-vehicle AR (partial), Dealer VR (preferred for simulation), Online Portal (didactic only)
Module 4 — Privacy and Data Rights	1 hour	Data categories, retention periods, access requests, deletion rights, opt-out options, third-party sharing, Data	80%	In-vehicle AR, Dealer VR, Online Portal

Module	Duration	Key Topics	Assessment Pass Threshold	Delivery Format Options
		Sovereignty Charter		
Module 5 — Certification Assessment	Variable (1–2 hours)	Comprehensive 50-question integrated assessment; practical competency demonstrations	85% (Class II/III); 75% (Class I)	In-vehicle AR (partial), Dealer VR (full with practical), Online Portal (written component only)

Appendix G — Manufacturer Compliance Checklist

This checklist is provided as a self-assessment tool for manufacturers. SVSC certification requires documented evidence of compliance with all items. This checklist does not substitute for formal SVSC TRB review.

36. ☐ PSA hardware installed meeting all specifications in Section 13.1 and Appendix D
37. ☐ Hot-standby backup sensors installed for all safety-critical PSA components (Section 13.2)
38. ☐ Failsafe default behaviors defined and tested for each sensor failure mode (Section 13.2)
39. ☐ Safe Harbor Navigation System hardware installed: 20% (Class I/II) or 25% (Class III) emergency propulsion reserve, auto-door lock on Hard Lock, external status light system, external communication interface (Section 13.3)
40. ☐ BRI algorithm documented and submitted to SVSC TRB for certification (Section 14.1)

41. ☐ BRI algorithm explainability requirement demonstrated (no black-box sole-reliance) (Section 14.1)
42. ☐ Whisper Course implementation submitted to SVSC Ethics Board for ethics certification (Section 14.2)
43. ☐ OTA update approval process established per Section 14.3; rollback capability validated
44. ☐ Data architecture: AES-256 encryption at rest and in transit; 90-day local retention; 2-year cloud retention; PP sovereignty tools in vehicle UI (Section 14.4)
45. ☐ Pre-market testing completed: 10,000 BRI simulations; Whisper ethics stress-test; Hard Lock reliability ≥99.99%; Safe Harbor accuracy ≤50m; ESC demographic accuracy ≥94% overall / ≥92% per group (Section 15.1)
46. ☐ PP onboarding training program available in all three delivery formats (Section 11.6)
47. ☐ Fleet Operator Certification program available (Section 12.1)
48. ☐ First Responder Integration Training made available at no cost (Section 12.2)
49. ☐ Manufacturer Service Ethics Training established; data chain-of-custody procedures documented (Section 12.3)
50. ☐ Quarterly compliance reporting system established (Section 15.2)
51. ☐ Annual third-party audit process established with SVSC-accredited auditor (Section 15.2)
52. ☐ Incident reporting system capable of meeting all tier reporting timelines (Section 18.2)
53. ☐ PP dispute resolution portal operational (Section 8.4)
54. ☐ ISO 26262 ASIL-D compliance for BRI co-processor and Hard Lock systems demonstrated
55. ☐ Annual bias audit process established; remediation process documented (Section 10.4)

Appendix H — Regulatory Cross-Reference Table

SVSC Doctrine Provision	FAA Regulation	NHTSA Regulation	ISO Standard	SAE Standard
Section 1.2 (Scope — Airspace)	14 CFR Parts 91, 107; FAA UAM Integration Framework	—	—	—
Section 1.2 (Scope — Terrestrial)	—	49 CFR Parts 571, 575 (FMVSS)	—	SAE J3016
Section 1.4 (Doctrine Hierarchy)	14 CFR Part 91 (general flight rules)	49 CFR Part 571 (FMVSS)	ISO 26262:2018	SAE J3016:2021
Section 13.1 (PSA Hardware — Encryption)	—	NHTSA Cybersecurity Best Practices for Motor Vehicles	ISO/SAE 21434:2021 (Automotive Cybersecurity)	SAE J3061
Section 13.2 (Redundancy — ASIL)	DO-178C (Software); DO-254 (Hardware)	—	ISO 26262:2018 ASIL-D	—
Section 13.3 (Safe Harbor — Class II/III)	14 CFR Part 77 (Obstructions); FAA AC 90-23 (Emergency Landings)	—	—	—
Section 14.3 (OTA Updates)	FAA AC 20-115 (Software Approval)	NHTSA AV 4.0 Guidance (OTA)	ISO 26262 Part 8 (Change Management)	—
Section 10.2 (PSA RF Sensors)	14 CFR Part 87 (RF Interference); RTCA DO-160	FCC Part 15 (Class B)	—	—

SVSC Doctrine Provision	FAA Regulation	NHTSA Regulation	ISO Standard	SAE Standard
Section 16.2 (International — EASA)	ICAO Doc 9854 (Global ATM Operational Concept)	—	—	—
Section 14.4 (Data Privacy)	—	—	ISO/IEC 27001:2022 (Information Security)	—

Appendix I — Full Glossary (Alphabetical)

This alphabetical glossary compiles all defined terms from Section 2.1 and terms introduced throughout the Doctrine, for comprehensive reference. Terms appearing in **bold** are primary defined terms; terms appearing in regular weight are supplementary references.

Term	Section Defined	Definition Summary
Agitation-on-Notification (AoN)	2.1	Behavioral pattern where BRI increases following explicit notification of course reassignment; primary Whisper Course activation criterion.
Air Corridor Suspension	5.3.2	Suspension of pending altitude transition during Naughty Course Phase 2; no airborne operations until BRI < 50.
Anti-Gaming Protection	7.4	Mechanism preventing artificial manipulation of BRI decay signals; detects deliberate biosignal manipulation and suspends decay augmentation.
Baseline Calibration	10.3	First 90 days of Space Car ownership during which the individual Peaceful

Term	Section Defined	Definition Summary
Period		Baseline is established.
Behavioral Risk Index (BRI)	2.1 / 7	Composite real-time score (0–100) representing PP behavioral risk level; drives all course assignments and threshold triggers.
BRI Decay	2.1 / 7.4	Natural or intervention-assisted reduction of BRI; natural rate: 2 points/minute under calm conditions.
BRI Hard Reset	7.4	Full reset of BRI to 0 following PP vehicle exit and 30-minute non-occupancy period.
Binaural Relaxation Frequencies	2.1 / 6.3.1	Low-frequency audio overlays (40–60 Hz) deployed covertly during Whisper Protocol to reduce physiological arousal.
Child-Alone Protocol	9.4.4	Specialized ESC-7 response for detection of child as sole vehicle occupant; prevents journey initiation; transmits to emergency contact and law enforcement.
Cognitive Load Sub-Index (CLI)	2.1 / 7.1.5	BRI sub-component (15% weight) measuring cognitive resource consumption via response latency and dual-task performance.
Compliant Fleet Operator	2.1	Commercial/governmental entity operating Space Car fleet with complete SVSC Fleet Operator Certification and reporting obligations.
Conversational Deflection	6.3.5	Whisper Protocol intervention in which AI assistant gradually redirects PP conversation from agitating to neutral topics.
Course Path	2.1	Operational behavioral management mode: Peaceful Course, Naughty Course, or Whisper Course.
Domestic Conflict Detection	9.4.3	ESC-7 sub-protocol for detecting intra-cabin domestic violence or coercive control patterns; triggers law enforcement dispatch.
Emotional State Classification (ESC)	2.1 / 9	Seven-state categorical model classifying PP emotional condition from ESC-1 (Serene) to ESC-7 (Crisis).

Term	Section Defined	Definition Summary
Environmental Sub-Index (ESI)	2.1 / 7.1.4	BRI sub-component (10% weight) derived from traffic density, weather, time of day, and geofence zone.
Ethics Board	16.3	SVSC governing body responsible for Whisper Course ethics certification, bias audit review, and PP rights policy.
Full System Override	8.2	Automated state at BRI 90–99; all PP inputs disabled; law enforcement notified; emergency services dispatched.
Geofence	2.1	Virtual geographic boundary enforcing modified vehicle behavior within a specified area; may be static or dynamic.
Good Faith Compliance Shield	17.2	Liability protection for manufacturers demonstrating documented good-faith compliance with this Doctrine at time of incident.
Hard Lock	2.1 / 8.2	Automated state at BRI 75; full autonomous control; PP navigation input disabled; EMS dispatched; vehicle sealed on Safe Harbor arrival.
Historical Risk Sub-Index (HRI)	2.1 / 7.1.3	BRI sub-component (15% weight) from 30-day BRI average, prior Naughty Course activations, and flag history.
Independent Review Panel (IRP)	18.3	SVSC-convened panel investigating Tier 3 and Tier 4 incidents; issues Finding Report within 60–90 days.
Individual Psychological Profile (IPP)	2.1 / 10.1	Persistent privacy-protected PP behavioral data record; stores Peaceful Baseline, trigger patterns, de-escalation preferences, BRI history.
Kinematic Sub-Index (KSI)	2.1 / 7.1.2	BRI sub-component (20% weight) from gesture velocity, input force, and body position shifts.
Lighting Gradient	6.3.3	Whisper Protocol intervention reducing cabin luminance and shifting color temperature toward calm-spectrum blue-green at imperceptible rates.
Maximum Danger State	8.2	BRI 100; forced Safe Harbor landing/parking; vehicle sealed; law enforcement notification with full data

Term	Section Defined	Definition Summary
		transmission.
Micro-Acceleration Smoothing	2.1 / 6.3.2	Whisper Protocol intervention reducing propulsion acceleration variance by 15–40% to decrease physiological arousal.
Naughty-Course Flag	2.1	Discrete behavioral trigger event activating Naughty Course regardless of current BRI score.
OTA Update	2.1 / 14.3	Remote software update to Space Car systems; safety-critical OTA requires SVSC TRB pre-approval.
PP Advocacy Panel	16.3	SVSC governing body handling PP dispute escalation and PP rights oversight.
Peaceful Baseline	2.1 / 10.3	Individually calibrated BRI and biometric baseline established during first 90 days of Space Car ownership.
Physiological Sub-Index (PSI)	2.1 / 7.1.1	BRI sub-component (30% weight) from HRV, GSR, micro-expression analysis, and vocal stress analysis.
Pilot-Passenger (PP)	2.1	Primary registered human occupant authorized to issue navigational commands and invoke manual override protocols.
Psychometric Sensor Array (PSA)	2.1 / 10.2	Integrated hardware suite capturing biometric, kinematic, visual, and acoustic data for BRI/ESC computation.
Resistant to Direct Intervention (RDI)	6.2	ESC flag: PP has rejected or escalated in response to three or more direct interventions in the current or recent journeys.
Route Softening	6.3.4	Whisper Protocol intervention gradually adjusting heading toward Safe Harbor via natural route complexity without explicit announcement.
Safe Harbor	2.1 / Appendix E	Designated pre-approved location with emergency services access, structural safety, and perimeter security for Hard Lock arrival.
Single-Event Override Flag (SEOF)	2.1 / 8.3	Discrete detection event triggering immediate Hard Lock regardless of BRI; includes weapon detection, medical emergency, child endangerment, direct

Term	Section Defined	Definition Summary	
		threat vocalization.	
Social Dynamics Sub-Index (SDI)	2.1 / 7.1.6	BRI sub-component (10% weight) from passenger count, detected verbal conflict, and vulnerable occupant flags.	
Soft Lock	2.1 / 5.3.3	Operational state at BRI 70–74; 25 mph maximum; Safe Harbor navigation; EMS pre-staged; cabin recording; PP notified.	
Space Car	2.1	Multi-modal autonomous/semi-autonomous vehicle capable of terrestrial roadway, low-altitude airspace, and/or transitional corridor operations.	
SVSC Technical Review Board (TRB)	2.1 / 16.3	SVSC body certifying BRI algorithms, reviewing OTA updates, and adjudicating manufacturer compliance disputes.	
	Threshold Override	2.1 / 8.1	Automated system action at a specific BRI threshold that cannot be countermanded by PP input without biometric re-authorization.
Three-Course Framework	3	Real-time psychological simulation engine continuously evaluating PP state and assigning one of three Course Paths.	
Whisper Protocol	2.1 / 6.3	Set of covert, non-declarative calming interventions applied during Whisper Course without PP awareness of course reassignment.	
Whisper Course	6	Covert de-escalation pathway embedded within Naughty Course; applies calming interventions transparently to the PP when direct notification would worsen behavioral state.	

Appendix J — SVSC Contact Information and Reporting Channels

All communications, submissions, and regulatory reports to the Space Vehicle Safety Consortium must be directed through the appropriate channel listed below. Misdirected submissions do not satisfy reporting obligations under this Doctrine.

Purpose	Channel	Contact / Portal	Response Time Commitment
BRI Algorithm Certification Submission	SVSC Technical Review Board (TRB) Portal	trb-submissions@svsc.gov (secure upload portal)	Acknowledgment within 5 business days; full review within 45 days
Whisper Course Ethics Certification	SVSC Ethics Board Portal	ethics-certification@svsc.gov	Acknowledgment within 5 business days; review panel convened within 30 days
OTA Update Pre-Approval (Standard)	SVSC TRB OTA Portal	ota-review@svsc.gov	Acknowledgment within 2 business days; decision within 45 days
OTA Emergency Patch Approval	SVSC Emergency Operations Desk	emergency-ops@svsc.gov | +1-800-SVSC-911 (24/7)	Emergency panel convened within 24 hours; decision within 72 hours
Tier 3 Incident Preliminary Notification	SVSC Incident Reporting Portal	incident-report@svsc.gov (secure; timestamped submission)	Automated acknowledgment within 1 hour; case assigned within 4 hours
Tier 4 Critical Event — Immediate Notification	SVSC Emergency Operations Desk	+1-800-SVSC-911 (24/7) AND critical-incident@svsc.gov	Immediate acknowledgment; Independent Review Panel convened within 48 hours
Quarterly Compliance Report Submission	SVSC Compliance Management System	compliance-portal.svsc.gov (manufacturer	Automated acknowledgment; review completed

Purpose	Channel	Contact / Portal	Response Time Commitment
		login required)	within 20 business days
Annual Third-Party Audit Report Submission	SVSC Compliance Management System	compliance-portal.svsc.gov (manufacturer login required)	Acknowledgment within 5 business days; TRB review within 45 days
PP Threshold Dispute (Escalated)	SVSC PP Advocacy Panel	pp-advocacy@svsc.gov | pp-portal.svsc.gov	Acknowledgment within 3 business days; case assignment within 10 business days
Bias Complaint (PP-Filed)	SVSC Ethics Board — Bias Complaint Channel	bias-complaint@svsc.gov (anonymous submission permitted)	Acknowledgment within 3 business days; Ethics Board review within 30 days
Safe Harbor Zone Designation Application	SVSC Safe Harbor Registry	safeharbor@svsc.gov | safeharbor-registry.svsc.gov	Acknowledgment within 5 business days; inspection scheduling within 30 days; decision within 60 days
Doctrine Amendment Proposal (Public)	SVSC Public Comment Portal	amendments.svsc.gov/propose	Acknowledgment within 5 business days; forwarded to Technical Committee for review
First Responder Training Enrollment	SVSC Training Portal	training.svsc.gov/firstresponder	Immediate access upon agency registration
General Public Inquiries	SVSC Public Information Office	info@svsc.gov | +1-800-SVSC-INFO (Mon–Fri, 8 AM–6 PM ET)	Response within 10 business days
Law Enforcement Real-Time Data Request (Active Incident)	SVSC Law Enforcement Liaison Desk	+1-800-SVSC-LEO (24/7) | Secure law enforcement portal: leo.svsc.gov	Immediate response for active incidents; credentialed law enforcement only
Media and Press Inquiries	SVSC Office of Communications	press@svsc.gov	Response within 2 business days for routine; 4 hours

Purpose	Channel	Contact / Portal	Response Time Commitment
			for incident-related press inquiries

ⓘ **Reporting Channels Note**

All contact information listed in this Appendix is maintained in the SVSC Contact Registry at

contacts.svsc.gov

. In the event of changes to contact information, the SVSC Contact Registry is the authoritative source. Manufacturers and operators are advised to verify contact details annually and to register their compliance officer contact information with the SVSC Compliance Management System to ensure timely receipt of regulatory communications.

Document Authority and Certification

This document — **Space Car Safety Doctrine — Unified Edition, Policy Version 1.0** (SVSC-DOC-2026-001) — has been reviewed, approved, and adopted by the Space Vehicle Safety Consortium Governing Board in accordance with the Amendment Process defined in Section 19.2.

This Doctrine is effective as of **01 January 2026** and supersedes all prior SVSC draft guidance documents, technical advisory notices, and pre-publication working drafts pertaining to Space Car behavioral safety. All certified manufacturers, registered fleet operators, and authorized dealers are bound by its provisions as a condition of their SVSC certification status.

Signatory	Role	Governing Body	Date of Approval
[SVSC Executive Director]	Executive Director	SVSC Governing Board	12 August 2025
[TRB Chair]	Chair, Technical Review Board	SVSC Technical Review Board	12 August 2025
[Ethics Board Chair]	Chair, Ethics Board	SVSC Ethics Board	12 August 2025
[PP Advocacy Panel Chair]	Chair, PP Advocacy Panel	SVSC PP Advocacy Panel	12 August 2025
[Law Enforcement Liaison Chair]	Chair, Law Enforcement Liaison	SVSC Law Enforcement Liaison	12 August 2025
[Manufacturer Liaison Committee Chair]	Chair, Manufacturer Liaison Committee	SVSC Manufacturer Liaison (Advisory)	12 August 2025

Governing Board Vote Record: Adopted by unanimous vote (6/6) of the SVSC Governing Board on 12 August 2025. No dissenting votes. No abstentions. Quorum confirmed (6/6 members present).

Public Comment Period: 60-day public comment period closed 15 July 2025. All 847 public comments received were reviewed and categorized. Material comments resulting in Doctrine revisions are documented in SVSC Comment Response Record SVSC-CRR-2025-001, available at amendments.svsc.gov.

Ethics Board Opinion: Ethics Board issued unanimous favorable opinion (7/7) on 28 July 2025, with six advisory recommendations incorporated into the final text (Sections 6.4, 10.4, 12.3, 14.2, 17.3, and Appendix C).

Technical Review Board Certification: TRB issued certification of technical feasibility and regulatory alignment (6/9 super-majority) on 5 August 2025. Three TRB members issued concurring opinions recommending enhanced sensor redundancy specifications for Class III vehicles, incorporated as Section 13.4.

■ Final Regulatory Note — Living Document Commitment

The SVSC commits to reviewing this Doctrine in full no later than

01 January 2027

— one year following the effective date — to incorporate operational experience from the first year of Space Car deployments under this framework. All manufacturers, fleet operators, PPs, first responders, and regulatory partners are invited to submit observations, performance data, and recommended improvements through the SVSC Public Comment Portal (amendments.svsc.gov) at any time. The improvement of this Doctrine is a shared public responsibility, and the SVSC welcomes every voice in that effort.